Seven ordered points and the nonexistence of arithmetic ball quotient compactifications

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Abstract

We prove that the ordered GIT quotient of seven points on $\mathbb{P}^1$ with equal weights is not isomorphic to the Baily–Borel compactification of any arithmetic complex ball quotient. Together with our companion paper and the known positive examples, this determines the equal weight case for every $n \geq 6$: such a realization exists precisely for $n = 6, 8$. The lattice and the algebraic isomorphism are unrestricted, and torsion and divisorial ramification are allowed. The proof combines arithmetic reductions with identities involving Chern classes of fixed subvarieties and intersection inequalities for nef divisors on the GIT quotient. The final comparison includes all fixed components and requires no bound on the degree of an individual fixed curve.

Use of AI. The authors used OpenAI Codex as a discussion partner for mathematical arguments, to assist with revising proofs and their presentation, and to proofread the manuscript. The authors take responsibility for the mathematics, computations, and references.

1 Introduction

Let

$$X = X_7 = (\mathbb{P}^1)^7 /\!/_{\mathcal{O}(1,\dots,1)} \mathrm{SL}(2,\mathbb{C})$$

be the ordered GIT quotient; the symmetric group is not divided out. The labels and their coincidence subvarieties remain part of its geometry.

Theorem 1.1. There is no arithmetic lattice $\Gamma \subset \mathrm{PU}(1,4)$ for which

$$X \simeq (\mathbb{B}^4/\Gamma)^{\mathrm{BB}}$$

as algebraic varieties.

The theorem allows arbitrary torsion, reflection ramification, and finite index. The isomorphism need not preserve the action on the labels, and the branch divisor is not prescribed. The passage to an integral model retains the original full group. A construction for a chosen subgroup or a cusp alone would not establish the theorem.

The companion paper [15] (version of October 16, 2026) proves the corresponding nonexistence for arbitrary positive rational weights on at least nine labels. For $n \geq 6$, let $X_n = (\mathbb{P}^1)^n /\!/_{\mathcal{O}(1,\dots,1)} \mathrm{SL}(2,\mathbb{C})$. Together with the known realizations at six and eight points, Theorem 1.1 yields the following classification.

Corollary 1.2. For every integer $n \geq 6$, the ordered equal weight GIT quotient X_n is the Baily–Borel compactification of an arithmetic ball quotient if and only if $n = 6$ or $n = 8$.

The positive realizations are the arithmetic Deligne–Mostow quotients [2, 8, 9]. Our contribution is the exclusion of all arithmetic realizations of X_7, rather than the failure of one period map. The global result is not a claim about nonarithmetic lattices or arbitrary compactifications.

The proof first excludes cocompact lattices and the nonuniform lattices over other fields. In the Gaussian case, the cusp geometry forces a norm condition on every actual reflection root. This condition produces a self-dual lattice preserved by the same full group. The branch character and the degrees on the thirty five coincidence planes then give $L - \frac{3}{2}H$ nef. No permutation of the labels is assumed to lift to the ball.

The final comparison uses every fixed surface, curve, interior point, and cusp. An average over the fixed sets gives a triangular system for the absolute Chern classes; its inverse is a binomial transformation. The determinant of the quotient differential gives a second relation along every fixed curve, with its full normalizer, finite characters, and actual cusp widths. The same local incidence equations apply to the projectivization at an interior point and the abelian quotient at a cusp. The nef cone of the configuration quotient is incompatible with these relations. We retain both root norms and all eight possible groups on curves throughout this final argument. No degree or index bound is imposed on an individual component. The common determinant formula is Proposition 3.1 of [15]; the absolute Chern identities and their application are proved here.

2 Geometry and preliminary comparisons

Put $P_7 = P = (\mathbb{P}^1)^7$, and let P^{s} be the locus where at most three labels coincide at any point. The effective diagonal action of $\mathrm{PGL}(2,\mathbb{C})$ is free on P^{s}, and stability equals semistability. The principal bundle $P^{\mathrm{s}} \to X$ defines

$$\pi_7 : \mathcal{P}_7 = (P^{\mathrm{s}} \times \mathbb{P}^1)/\mathrm{PGL}(2,\mathbb{C}) \longrightarrow X, \qquad [(p,z)] \longmapsto [p],$$

with sections $\sigma_i([p]) = [(p, p_i)]$. We use projective bundles of one dimensional quotients. The geometry of the quotient gives

$$H = -K_X = (2,\dots,2), \qquad \mathrm{Pic}(X) = \{b \in \mathbb{Z}^7 : \textstyle\sum_i b_i \in 2\mathbb{Z}\}.$$

For a set I of three labels, the coincidence subvariety Z_I is $\mathbb{P}^2$, its normal bundle is $\mathcal{O}_{\mathbb{P}^2}(-1)^{\oplus 2}$, and $H|_{Z_I} = \mathcal{O}_{\mathbb{P}^2}(1)$. There are thirty five such planes. These assertions follow from the free quotient and its marked sections, as in [15, Proposition 2.5], [15, Equation 2.2], and [15, Proposition 2.6].

Assume a hypothetical isomorphism in Theorem 1.1, and identify its ball quotient with the open part of X. Use the model of negative lines in a Hermitian space of signature $(1,4)$. Let $\mathcal{L}$ be its automorphic rational line bundle, normalized by $K_{\mathbb{B}^4} = \mathcal{L}^5$, and put $L = c_1(\mathcal{L})$. For a full normalizer curve $f : C \to X$, set

$$a = \deg f^*H, \qquad t = \deg f^*L, \qquad x = g(C) - 1.$$

All finite inertia characters are retained in the descended bundles E_e associated with an invariant of polynomial degree e; their numerical degrees are $-et$ before the stated local corrections.

We use [15, Proposition 2.2, Proposition 2.3, Proposition 2.4] for tangent quotients. For a negative target, every homomorphism factors through differences of coincident sections. If m labels coincide and their common relative tangent line has degree $-e < 0$, then

$$2 \le m \le 3, \qquad a = (7-2m)e + 2\delta, \qquad \delta \ge 0.$$

The remaining relative tangent lines generate the kernel and give the exterior power bounds of [15, Proposition 2.3]. A quotient of rank 2 is always required to be surjective at every point, not just at the generic point. The nef criterion of [15, Proposition 2.6] must be checked on every Z_I; it does not identify the ball structures along different planes.

The original full arithmetic group and its normalized lift are those of [15, Proposition 2.11]. For each fixed ball we use its full normalizer, the finite morphism of [15, Lemma 2.12], and the same restricted automorphic bundle. At a cusp, P denotes the full stabilizer, N its Heisenberg lattice, and $F = P/N$ its finite horizontal group. The horizontal torus A and its positive line bundle are defined with the least positive central period of this N, with the actual action on the fiber. For a smooth cusp, the horizontal holonomy is irreducible. For its actual linearization, the invariant sections form a polynomial ring, whose projectivization is generally a weighted projective space. The least central period gives its faithful linearization and the grading of its invariant section ring, as in [15, Lemma 2.16]. The lift fixing the point in [15, Lemma 2.17] applies to the kernel of the actual fiber character. The degree of an invariant section in this grading is distinct from the polynomial degree e of an invariant of the normal representation. In a fixed curve coordinate the ambient central coordinate satisfies $Q = q^{\kappa}$, where κ is its actual relative width. We use [15, Proposition 3.4] and [15, Lemma 3.5] only after specifying the larger stabilizers, end groups, characters, and width.

Lemma 2.1. For $X = X_7$, let $\mathcal{A}$ be the adjoint bundle of the principal bundle $P_7^{\mathrm{s}} \to X$, with fiber $\mathfrak{sl}_2(\mathbb{C})$. Put $\psi_i = c_1(\sigma_i^* T_{\mathcal{P}_7/X})$ and $p = -c_2(\mathcal{A})$. Then, in the rational Chow ring of X,

$$H = \sum_i \psi_i, \qquad \psi_i^2 = p, \qquad c_2(T_X) = \frac{H^2 - 5p}{2}, \qquad c_3(T_X) = \frac{H^3 - 13Hp}{6}.$$

Proof. The section σ_1 represents the universal family as $\mathbb{P}_X(\mathcal{V})$, where $\mathcal{V} = \pi_{7*}\mathcal{O}_{\mathcal{P}_7}(\sigma_1(X))$ has rank 2. For each quotient $\mathcal{V} \twoheadrightarrow \mathcal{L}_i$, the relative tangent line has first Chern class $2c_1(\mathcal{L}_i) - c_1(\mathcal{V})$, whose square equals $c_1(\mathcal{V})^2 - 4c_2(\mathcal{V}) = -c_2(\mathrm{End}_0(\mathcal{V})) = p$. The tangent sequence is $0 \to \mathcal{A} \to \bigoplus_i \sigma_i^* T_{\mathcal{P}_7/X} \to T_X \to 0$. Since $c_1(\mathcal{A}) = c_3(\mathcal{A}) = 0$, expanding $c(T_X) = \prod_i (1 + \psi_i)/(1 - p)$ through degree 3 proves the formulas. □

3 The automorphic degree on the coincidence planes

Throughout this section, $X = X_7$, $H = -K_X$, and $\mathcal{L}$ is the automorphic line bundle. For a curve $f : C \to X$, we write $a = \deg f^* H$ and $t = \deg f^* \mathcal{L}$. The degree of $\mathcal{L}$ is understood on the orbifold until an indicated power descends to X. For a fixed curve, $\mathcal{N}$ denotes $f^* T_X / (df(T_C))^{\mathrm{sat}}$. For an invariant direction of polynomial degree e preserved by the full normalizer, E_e denotes its line bundle with the finite inertia characters included; its degree is stated in each calculation. In the tables, a string such as 1112 abbreviates the tuple $(1, 1, 1, 2)$.

Lemma 3.1. Suppose that X is the Baily–Borel compactification of an arithmetic ball quotient, and let $Z \simeq \mathbb{P}^2$ be a three point coincidence plane. Write $\mathcal{L}|_Z = \lambda \mathcal{O}_Z(1)$. If the nontrivial part of the normal representation of the full generic stabilizer has rank 0 or 1, then $\lambda \geq 1$. If it has rank 2 and its least positive invariant degree is q, then $q\lambda \geq 1$.

Proof. On a normal neat cover, resolve a component above Z, with maps $h : S \to Y$ and $\tau : S \to Z$. The normalized component is finite Galois over the punctured plane, by transitivity of its component stabilizer on a general fiber. Let R be the common divisor of the maximal minors of dh. After removing R, the Gauss equation gives a positive current in $c_1(K_S - R - 3h^*\mathcal{L})$, whose potential is locally bounded above. On the inverse image of degree e of a general line avoiding cusps and exceptional images,

$$(K_S - R) \cdot \tau^* \mathcal{O}_Z(1) \geq 3\lambda e. \tag{3.1}$$

For ranks $0, 1$, apply [15, Lemma 3.10] with $k = c = 2$. The normal bundle is $\mathcal{O}_Z(-1)^{\oplus 2}$, the omitted cusp set is finite, and the full fixed normalizer retains the restricted $\mathcal{L}$ by [15, Lemma 2.12]. Thus $\lambda \geq 1$.

For rank 2, the fixed ball maps finitely to the punctured plane. Its normal bundle on a neat cover is $\mathcal{L}^{-1} \otimes \underline{U}$, with finite monodromy [15, Lemma 2.12]. If k independent invariants have the least positive degree q, the first normal Taylor coefficients of the plane ideal give a map $\mathcal{L}^{-q} \otimes \underline{U}_q \to \tau^* \mathcal{O}_Z(-1)^{\oplus 2}$ of generic rank k. Kill the finite monodromy on a compact inverse line; a negative coefficient degree would make this map zero. Thus $q\lambda \geq 1$. □

The following two assertions concern a nonuniform arithmetic realization over an imaginary quadratic field.

Lemma 3.2. Let $Z \simeq \mathbb{P}^2$ be any three point coincidence plane and put $\mathcal{L}|_Z = \lambda \mathcal{O}_Z(1)$. If the normal representation of its full generic stabilizer has rank 0, then $\lambda \geq 1$. If it has rank 1 and the generic reflection has order b, then $(1 - b^{-1})\lambda \geq 1/3$. If it has rank 2 and the least positive invariant degree is q, then $q\lambda > 1$. In all cases $w_K \mathcal{L} - 2H$ is nef, and hence $a \leq w_K t/2$ on every curve.

Proof. By Lemma 3.1, only equality $q\lambda = 1$ in rank 2 remains. Retain its notation and let c be the order of the full scalar kernel, so $\mathcal{L}^c$ descends and $q \mid c$. The bundle $\mathcal{S} = \mathcal{L}^q \otimes \tau^* \mathcal{O}_Z(-1)$ is torsion, with an equivariant trivialization of its cth power. The bundle $\mathcal{F} = \mathcal{S}^{-1} \otimes \underline{U}_q$ is finite flat. The central scalar ζ acts by ζ^q on both factors before taking their quotient; thus $\mathcal{F}$ and the map $\mathcal{F} \to \mathcal{O}_M^{\oplus 2}$ are equivariant for the actual projective component stabilizer. Constancy on compact inverse lines after a finite cover gives rank k at every interior point; the injective source then has trivial point inertia. It descends to the simply connected punctured plane as a trivial finite flat bundle, and Hartogs's theorem extends its constant rank k matrix to Z. [15, Lemma 2.15] supplies an actual cusp of this fixed ball. In a unipotent cover with full central coordinate Q and horizontal normal coordinates $y \in U$, the plane equations and their Taylor coefficients of degree q are divisible by Q, since the zero section contracts to the cusp. The trivialization of $\mathcal{S}^c$ is pulled back from $(\mathcal{L}^c|_Z)^q \simeq \mathcal{O}_Z(c)$ on the whole plane, using the Baily–Borel extension of [15, Proposition 2.11]. In the normalized Siegel frame and a regular frame of $\mathcal{O}_Z(-1)$, this trivialization is a unit at $Q = 0$. After killing the finite coefficient action on the same cover, its local cth root is a unit, so the direct Taylor coefficient matrix still vanishes at $Q = 0$. The constant rank $k > 0$ matrix therefore vanishes along a vertical ray, a contradiction.

Finally $q \leq w_K$ in every possible rank 2 group. For monomial groups this follows from their degrees $m, 2m/p$ and their character fields; a dihedral group has degree 2. For primitive groups, transfer to an actual cusp excludes G_5, G_8, G_{12}: the rank 3 primitive groups over quadratic fields are G_{24}, G_{25}, G_{26} and their only primitive parabolic group of rank 2 is G_4, of degrees $4, 6$ [7, Table 1], [10, Table 4]. Thus the remaining primitive case has $K = \mathbb{Q}(\omega)$ and $q = 4 \leq 6$. The integer $w_K \lambda$ is strictly greater than 1 in rank 2 and at least 2 in ranks 0 and 1. [15, Proposition 2.6] applied to the 35 planes gives the asserted nef inequality.

□

Lemma 3.3. In the setting of Lemma 3.2, if the generic stabilizer has rank 0 or 1, then $\lambda > 1$.

Proof. Suppose $\lambda = 1$ in rank 0 or 1. The equality case of [15, Lemma 3.10], with the same $k = c = 2$, makes the regular inverse image locally a totally geodesic ball of dimension 2, inside the generic fixed ball in rank 1. Let $B = \mathbb{B}(W)$ be this ball in the universal cover. The identity theorem makes B an entire irreducible component of the closed inverse image of Z. Local finiteness of the inverse components proves properness of its stabilizer quotient M_0; its generically injective immersion thus normalizes the component and maps finitely to the punctured plane. Remove deep cusp neighborhoods with compact complement [6]. Their intersections with B are horoballs if the isotropic vector lies in W, and bounded geodesic balls or empty otherwise [16]; in the latter case $h(\rho_W, \rho_W) = -h(\rho_U, \rho_U) < 0$ and the height is $-h(z, z)/|h(\rho_W, z)|^2$. Properness over a compact regular horosphere gives finitely many cuts,

with compact horizontal cross sections for the horoballs. The radial integral $\int_{u_0}^{\infty} u^{-3}\,du$, the bounded cuts, and the compact remainder prove finite volume.

Borel density makes the stabilizer on W irreducible [1]. The K algebra of its actual linear lifts is semisimple, since its complexification acts faithfully and completely reducibly on $W \oplus W^{\perp}$. Its unique central idempotent of rank 3 projects onto W; the positive complement has dimension 2, so Galois invariance of that rank defines both summands over K. The full preserved lattice makes the normal action finite, and the original full stabilizer gives a finite birational map onto the normal punctured plane, hence an isomorphism.

For rank 1, let U_0 be the normal axis fixed by the generic cyclic group; for rank 0 let it be the entire positive normal space. The first Taylor coefficients of the plane ideal give a generically injective map $\mathcal{L}^{-1} \otimes \underline{U}_0 \to \tau^{*}\mathcal{O}_Z(-1)^{\oplus 2}$. Apply the argument excluding equality in Lemma 3.2 with $q = 1$ and $\underline{U}_q = \underline{U}_0$. Let c be the order of the full scalar kernel and put $\mathcal{S} = \mathcal{L} \otimes \tau^{*}\mathcal{O}_Z(-1)$. The fixed ball is now defined over K, its full normal action is finite, and the two factors have the same central scalar character. The first derivatives also exclude an additional inverse branch by the implicit function theorem. Thus their quotient descends to a nonzero constant matrix on the punctured plane. Meyer in six variables supplies an actual cusp, and the same trivialization of $\mathcal{S}^{c}$ on the whole plane is a unit in the normalized Siegel frame. The direct coefficients are divisible by the full central coordinate Q, giving the same contradiction.

□

A coincidence plane with generic stabilizer of rank 2 is the compactification of its fixed ball quotient: [15, Lemma 2.12] gives a finite birational map, and the plane is normal.

4 Seven points and cocompact lattices

Throughout this section we assume that $X = X_7$ is the quotient of $\mathbb{B}^4$ by a cocompact arithmetic lattice Γ. We retain $H = -K_X$ and $L = c_1(\mathcal{L})$ with $K_{\mathbb{B}^4} = \mathcal{L}^5$. The branch divisor, its indices, and the isomorphism with X are arbitrary. [15, Proposition 2.11] gives a normalized realization generated by reflections over a CM field K; let F be its maximal real subfield and D_K, D_F their absolute discriminants. Every stabilizer below is the full stabilizer, and every quotient of a fixed ball uses the full normalizer. For a fixed curve $f : C \to X$ we put $x = g(C) - 1$, $a = \deg f^{*}H$, and $t = \deg f^{*}L$. When df vanishes on a divisor D, we use $N = f^{*}T_X/T_C(D)$.

4.1 Finite stabilizers

The parabolic classifications [7, Table 1], [10, Theorem 3.6 and Tables 5–9] give the following irreducible upper groups; products $T \times \mu_b$ retain every positive integer b before reflection orders are restricted.

T	irreducible upper groups	effective axis orders
$G(m,p,3),\ m \geq 6$	$G(m,p,4)$	m
$G(m,1,3),\ 3 \leq m \leq 5$	$G(m,1,4)$	m
$G(5,5,3)$	$G(5,5,4)$	5
$G(3,3,3)$	$G(3,3,4), G(3,1,4)$	$3,3$
B_3	B_4, F_4, G_{29}	$2,2,4$
$G(4,2,3), G(4,4,3)$	$G(4,2,4), G_{31};\ G(4,4,4), G_{29}$	4
G_{25}, H_3	$G_{32};\ H_4$	$6;\ 2$
A_3	$A_4, D_4, F_4, H_4, G_{29}, G_{31}$	

The full rank 4 list is A_4, $G(m,p,4)$, $F_4, H_4, G_{29}, G_{31}, G_{32}$. After Lemma 4.2, products of rank 3 are $T \times \mu_2$, and also $G_{25} \times \mu_3$; the rank 2 factors are products, $A_2, B_2, I_2(5), G(4,2,2), G_4$.

A factor with reflection orders $2,3$ cannot enter rank 4; on a fixed plane it is $G(6,2,2)$. For $G(m,p,2)$, the invariants $S = x^m + y^m$, $P = (xy)^{m/p}$ give branch equation $S^2 - 4P^p = 0$ and, for $p < m$, $P = 0$ of index m/p. For a mirror of index b and orbit size ρ, the effective normalized order is $|J|/(b\rho)$, separately on every branch of a self-intersection [28, Table 1]. The effective normalized orders for $A_2, I_2(5), B_2, G(4,2,2), G(6,2,2), G_4$ are $1,1,2,4,6,2$. The groups $A_2, I_2(5), G_4$ give cusps $(2,3),(2,5),(2,3)$; B_2 gives two branches of contact 2, and $G(4,2,2), G(6,2,2)$ give ordinary triple points. A product $\mu_b \times \mu_c$ gives a node of normalized orders c, b. Below plane degree $1/2$, the only irreducible upper pairs are $(G(4,2,2), G(4,2,3))$, (G_4, G_{25}), (μ_3^2, G_{25}). Their character corrections and normal determinant orders are $(1/2,2)$, a nonimmersed axis (v^2, v^3), and $(1/2,3)$, respectively. The further product $G_4 \times \mu_3$ for μ_3^2 has the same nonimmersed cusp. At G_{32} the effective plane group $G(6,1,2)$ has degrees $6, 12$; ambient degrees $12, 18, 24, 30$ omit its degree 6 cotangent direction. Thus this point is excluded from a smooth fixed plane, and any absorbed cyclic factor meets an excluded upper locus. This proves the product assertion used in Lemma 4.3.

4.1.1 Monomial groups

Put $k = mt$, $D = 3m + 3m/p$ on an ordinary monomial axis. The degree m direction is unique except for $p = 3$. For $m \geq 6, p = 3$, coordinate mirrors have squared normalized inner products in $\{0, 1/2\}$, but diagonal mirrors have a pair with value $\cos^2(\pi/m) \notin \{0, 1/2\}$. Thus the full normalizer preserves the coordinate mirrors and their elementary symmetric invariant of degree m. The character at every larger monomial group and product is trivial, so $k \in \mathbb{Z}_{>0}$. The curve equations [15, Proposition 3.4] are $2x + (1 - m^{-1})R + \sigma = 2t$ and $a = -2x - Dt + (2 + p^{-1})R$, where $\sigma = 0$ or $\sigma \geq 1/2$. For $m \geq 6$, and for $3 \leq m \leq 5, p = 1$, the inequality $D > 2(2 + p^{-1})/(1 - m^{-1})$ gives $g = 0$. The finite test uses the following table; h is the target associated with the invariant of greatest degree degree.

Group	Larger group	h	Finite inequalities or surviving row
$G(m,1,3)$, $m \geq 6$	$G(m,1,4)$	$R - 3k$	$2k \leq R \leq 2(k+m)/(m-1)$; $k(m-2) \leq m$.
$G(m,p,3)$, $m \geq 6, p > 1$	$G(m,p,4)$		$R > (3k-2)/2$; $k < (6m-2)/(3m-7)$.
$G(5,5,3)$	$G(5,5,4)$	$-3t + R/5$	$(k,R,a,t) = (1,3,5,1/5)$.
$G(m,1,3)$, $3 \leq m \leq 5$	$G(m,1,4)$	$R - 3k$	$1 \leq k \leq \lfloor m/(m-2) \rfloor$; $2k \leq R \leq \lfloor 2(k+m)/(m-1) \rfloor$; $a = 2 - 6k + 3R$.

In the first row $(k,R,a,h) = (1,2,2,-1)$ contradicts $a \geq 3$. In the second row $k \leq 2$, except $m = 6, k = 3$ where $a \leq 0$; substituting $k = 1, 2$ leaves no positive integral a with $\sigma = 0$ or $\sigma \geq 1/2$. For the other rows, [15, Lemma 3.5] applied to coefficient degrees $R - 2k, R - k, R$ and all R upper zeros gives Hom dimension 1, attained by the projection of the inverse differential to the coordinate of greatest invariant degree; at $G(5,5,4)$ the other degrees are $1, 2 < R$. The GIT tangent sequence excludes degree 0 for odd a and degree 1 for even a: negative relative tangent degrees contribute at least $2, 4$, respectively, and otherwise projective evaluation gives twice an integer. Together with $a \geq -3h$ for $h < 0$, this excludes $G(5,5,4)$ and leaves only $(m,k,R,a,t,\sigma) = (3,2,5,5,2/3,0)$, of relative tangent degrees $(-1,-1,3,1,1,1,1)$.

Lemma 4.1. For an ample GIT class of multidegree (b_i) and half sum q, one has

$$D^4 = 3q^4 - 2\sum_i (q - b_i)^4 + \sum_{i<j} (q - b_i - b_j)^4.$$

If its degree on a curve of relative tangent degrees $(-1,-1,3,1,1,1,1)$ is $2/3$, then $D^4 \leq 28/81$. For any cocompact arithmetic lattice of definition field K/F, put $h_{K,5} = |\operatorname{Cl}(K)[5]|$. Then

$$\chi_{\mathrm{orb}} \geq \frac{5}{h_{K,5}} D_F^{12} (D_K/D_F^2)^7 \left(\frac{288}{(2\pi)^{14}} \right)^{[F:\mathbb{Q}]}.$$

Proof. Extract the coefficient of k^4 in $\sum_{|I|\le 3}(-1)^{|I|}\binom{k(q-\sum_{i\in I}b_i)+5-|I|}{5}$. Concavity of $(D^4)^{1/4}$ permits averaging to (x,x,z,y,y,y,y), where $r_0 = x-y$, $y = 2/3 + r_0 - 3z/2$, $x = 2/3 + 2r_0 - 3z/2$, $0 < z < 1/3$, $|r_0| < z/2$. For $u = 3z$, $v = r_0/z + 1/2$, the coefficients of $648D^4$ in $B_i(u)B_j(v)$, $B_j(T) = \binom{4}{j}T^j(1-T)^{4-j}$, have maximum 224 at $(i,j) = (2,4)$. This nonnegative basis sums to 1, giving $D^4 \le 28/81$. For split nonhyperspecial places S_{split}, the maximal normalizer index is at most $5^{1+|S_{\mathrm{split}}|}h_{K,5}$ and each local factor exceeds 5. Cancel these factors in Prasad's formula, retaining factors 5 from the compact dual and central covering [20, Sections 2.3 and 2.5–2.9, equations (3), (5), (6), (8)]. □

The integral ample class cL, for scalar order c in [15, Proposition 2.11], has positive integral degree on every coincidence line. Complementary triples therefore give $cL - H$ nonnegative multidegrees and triple degrees; [15, Proposition 2.6] yields $a \le ct$. For the last $G(3,1,4)$ row the fixed signature is $(0;3,3,3,3,3,3)$, of area $4/3$. Odd generators lie in an Eichler norm one group of genus zero because the normalizer quotient is an elementary abelian 2-group [11, Theorem 2.1 and Section 3]. Voight's exhaustive 1209 groups of genus at most 2 give $[F:\mathbb{Q}] \le 7$ [30, Theorem in the Introduction and Section 4]. Here $c \ge 8$, $\zeta_3, \zeta_c \in K$, and $m = \mathrm{lcm}(3,c)$ lies in $\{3,6,9,12,15,18,21,24,30,36,42\}$. For each compact genus zero row of area μ, require integral $N = (4/3)/\mu$, an embedding $\mathbb{Q}(\zeta_m + \zeta_m^{-1}) \hookrightarrow F$, and cycles $N = ex_e + (e/3)y_e$ at each cone point, with $x_e, y_e \ge 0$, $y_e = 0$ if $3 \nmid e$, and $\sum y_e = 5$. Testing embeddings by defining polynomials gives the following rows. The rows $(D_F, \text{signature}, N)$ are $(12,(0;3,3,6),8)$, $(81,(0;2,3,9),24)$, $(81,(0;3,3,9),6)$ and $(1125,(0;3,3,15),5)$. The field polynomials are T^2-3, T^3-3T+1, $T^4-T^3-4T^2+4T+1$, giving $K = \mathbb{Q}(\zeta_{12}), \mathbb{Q}(\zeta_9), \mathbb{Q}(\zeta_{15})$. In the first field $c = 12$; traces contain every reflection root of unity, but an order 4 or 12 reflection would remain in a factor of rank 2 with equal degrees $q \mid 12$, where no such reflection exists. All orders therefore divide 6, and the generated determinant lies in μ_6, giving $c \mid 30$, a contradiction. The other two fields have class number 1; Lemma 4.1 gives lower bounds greater than $6,270$, contrary to $\chi_{\mathrm{orb}} = 5L^4 \le 140/81$.

4.1.2 The group $G(3,3,4)$

After excluding $G(3,1,4)$, the $G(3,3,3)$ axis has only $G(3,3,4)$ and product upper groups. The normalizer preserves degree 6, but may mix the degree 3 directions. The curve equations give $k = 6t \in \mathbb{Z}_{>0}$, $g = 0$, $k \le 10$, $a = 2 - 2k + 7R/3$, $\sigma = 2 + k/3 - 2R/3$, leaving exactly

$$(k,R,a,\sigma,h) = (2,3,5,2/3,1),(3,3,3,1,0),(4,3,1,4/3,-1),(6,6,4,0,0), \quad h = R-k.$$

The projection of the inverse differential to the coordinate of greatest invariant degree is everywhere surjective onto $\mathcal{O}(h)$. The invariants $v+S_1, vS_1+S_2, vS_2+P^3, uP$, with $v = u^3$, $S_1 = \sum y_j^3$, $S_2 = \sum_{i<j} y_i^3 y_j^3$, $P = y_1y_2y_3$, give $dS_1 = dg_2/v$, $dP = dg_4/u$ on its kernel. Hence every kernel line subbundle has degree at most $R - k/2$ after killing finite monodromy, without assuming a splitting on C. For $k = 3$, odd relative tangent degrees of sum 3 force negative labels: three give degree $7 > 3/2$ in the kernel, two prevent surjection to $\mathcal{O}$, and one requires $a \ge 5e$.

Apply the preceding cycle conditions to the compact genus zero rows in Voight's table, with area $2/3$ and four order 3 points, or area 2 and six order 3 points. The second test gives the fields below; each pair $([F:\mathbb{Q}], D_F)$ identifies a unique defining polynomial in that table.

$[F:\mathbb{Q}]$	D_F
1	1
2	5, 8, 12, 13, 17, 24, 28, 29, 33, 60
3	49, 81, 148, 169, 229, 257, 316, 321, 361, 469
4	725, 1957, 2000, 2304, 2777, 3981, 4205, 4352
5	24217, 36497, 38569

The first test deletes $24, 29, 33, 60, 321, 361, 469, 2304, 3981, 4205$, leaving 22 fields. For either test, N is the required area divided by the area of the tabulated row, and each cone order e permits precisely the integers $0 \le y_e \le 4$ or $0 \le y_e \le 6$ satisfying $N - (e/3)y_e \in e\mathbb{Z}_{\ge 0}$ and $y_e = 0$ if $3 \nmid e$; the sum of the y_e is 4 or 6. These bounded conditions specify every row used in the test. For $k = 2$, use the first list. The conditions $c \ge 15$, $K = F(\zeta_3)$, $F \ne \mathbb{Q}$, $|\mu(K)| \ge 15$ leave only $K = \mathbb{Q}(\zeta_9)$, $D_F = 81$. Three negative labels violate the kernel bound $13 > 2$; otherwise Lemma 4.1 for $2L$, or averaging with $y = 1/9 + z/6$, $0 < z < 1/3$, gives $L^4 \le 7/324$ or $31/1944$. Every ideal class of $\mathbb{Q}(\zeta_9)$ has a representative of norm at most 5 by Minkowski's theorem. The primes $2, 5$ have residue degree 6, and the unique prime above 3 is generated by $1-\zeta_9$; hence the class number is 1. Both contradict the arithmetic bound greater than 6. For $k = 6$ the kernel bound excludes negative degrees by $10 > 3$; immersion then gives $N = \mathcal{O}^2 \oplus \mathcal{O}(2)$, $t = 1$. Average to $(z, z, z, z, z, 1/2, 1/2)$, $1/3 < z < 1$, giving $L^4 \le 27/8$. The same filters with area 2 and six order 3 cycles give 32 polynomials. For $K = F(\zeta_3)$ put $d = [F : \mathbb{Q}]$, $\Delta = D_K/D_F^2$, $c_0 = 288/(44/7)^{14}$, and

$$M_K = \left\lfloor \left(\frac{4}{3}\right)^d \frac{(2d)!}{(2d)^{2d}} (\lfloor \sqrt{D_K} \rfloor + 1) \right\rfloor, \qquad U_K = \sum_{1 \le m \le M_K} \prod_{p^a \| m} \binom{2d + a - 1}{a}.$$

Minkowski's theorem and the coefficients of ζ_K give $h_{K,5} \le h_K \le U_K$. Every ideal class has a representative of norm at most M_K, and the number of ideals of norm p^a is at most $\binom{2d+a-1}{a}$. For the listed fields, $\Delta = 3^d$ except that $\Delta = 1$ at $D_F = 12, 24, 33, 60, 2304$, $\Delta = 3$ at $81, 321$, and $\Delta = 9$ at 3981. The lower bound $5D_F^{12}\Delta^7 c_0^d / U_K$ exceeds $135/8$ in the following cases; the last column bounds its integer part from below for every indicated field.

d	D_F	lower bound
2	$13, 17, 28, 29, 60$	52
3	$49, 148, 169, 229, 257, 316, 321, 361, 469$	443
4	$725, 1957, 2000, 2304, 2777, 3981, 4205, 4352$	1343
5	$24217, 36497, 38569$	10^{19}

.

For $D_F = 33$, one has $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ and $M_K = 5$. The prime 2 has two primes of residue degree 2, the prime 3 has two primes of ramification index 2 and residue degree 1, and the prime 5 has residue degree 2. Thus the prime ideals of norm at most 5 have norm 3 or 4. Each prime of norm 4 is principal, generated by a conjugate of $(\sqrt{-3}+\sqrt{-11})/2$, whose norm is 4. The square of each prime of norm 3 is principal, generated by a conjugate of $(1+\sqrt{-11})/2$. Thus $h_{K,5} = 1$, and the volume lower bound exceeds $30 > 135/8$. Consequently $F \ne \mathbb{Q}$, $c \ge 4$ and the volume bound leave

D_F	D_K	$\lvert\mu(K)\rvert$	$\rho_{K/F}$
5	225	6	31/6075
8	576	6	201641/124416
12	144	12	23/497664
24	576	6	551/2560
81	19683	18	18628987/12597120

.

The numerator divides $(cL)^4$ and is coprime to $5c$ because the maximal normalizer index is a power of 5. For $D_F = 81$ it exceeds $18^4(27/8) = 354294$. Otherwise $c = 6$, also using $c \mid 30$ for $D_F = 12$; positive triple degrees and moving weight sum 6 leave $(2, 3, 3, 3, 3; 3, 3)$, $(3, 3, 3, 3, 4; 3, 3)$, $(4, 4, 4, 4, 4; 2, 4)$, $(4, 4, 4, 4, 4; 3, 3)$, of volumes $586, 880, 1450, 1684$. None has the required divisibility, with special values from Section 4.7.1.

For $k = 4$ the negative target gives a line $C \subset Z$, $N = \mathcal{O}(1) \oplus \mathcal{O}(-1)^2$, and kernel of the projection to the coordinate of greatest invariant degree $\mathcal{O}(1) \oplus \mathcal{O}(-1)$. The degree 3 orbifold

bundle twisted by $\mathcal{O}_C(2)$ is unitary flat with coarse image $\mathcal{O} \oplus \mathcal{O}(-2)$; Bochner makes the trivial summand parallel. The characters $(1, \zeta_3^2)$, trivial product action and parallel normalizer invariant lines $P(S_1^3 - 27P^3) = 0$ select an S_3 fixed plane. Its compact quotient Y has maps $(U, V) \mapsto (U, V, 0, 0)$, $U = u^3 + z^3$, $V = u^3 z^3$, at upper points, and $(u, z) \mapsto (u^b, z^3)$ at stabilizers $T \times \mu_b$. Hence $N_{C/Y} = \mathcal{O}(-2)(D_R) = \mathcal{O}(1)$; the vanishing maps $\mathcal{O}_C(1)^2 \to \mathcal{O}_C(-j)$ and Krull intersection give $Y = Z$, $\lambda_Z = 2/3$. The product contribution $4/3$ and three index 2 contacts with C leave $\Delta_Z = 2C/3 + D/2 + 2E/3$, $(\deg D, \deg E) = (6, 2)$, or $\Delta_Z = 2C/3 + D/2 + 5A/6$, $(\deg D, \deg A) = (7, 1)$. The general groups are $G(3, 3, 3)$ on C and $S_3 \times \mu_b$ elsewhere. The full normalizer test [10, Theorem 3.12 and Tables 5–9] excludes A_4, B_4 by effective degrees $(1, 2), (2, 2)$ and $D_4, G(4, 2, 4), G(4, 4, 4)$ by normalizer/reflection orders $(2, 1), (8, 4), (4, 1)$; the same defect excludes G_{29}, G_{31}. For H_4, degree 6 of $I_2(6)$ is absent from $2, 12, 20, 30$; G_{32} lacks order 2 reflections. The S_3 types in $G(3, 3, 4)$ have degrees $(1, 3)$, excluded, and $(3, 6)$, giving the three points on C; in F_4, containment in B_3 contradicts general group $S_3 \times \mu_2$. Any rank 3 factor absorbing S_3 must act on its fixed line, since otherwise immersion requires degree 1; it therefore occurs only on C as $G(3, 3, 3)$. Off C the mixed factors in $S_3 \times J$ are $G(6, 2, 2), G_7$ in the first divisor and $G(12, 2, 2)$ in the second, of effective orders $6, 12, 12$. For A, Bézout and proportionality give $m + 2k = 7$, $-2 + 2/3 + m/2 + 11k/12 = 4/3$, hence $(k, m) = (10, -13)$. For irreducible E they give $m + 2k = 12$, $-2 + 4/3 + m/2 + 5k/6 = 8/3$, hence $(k, m) = (16, -20)$. If E splits, its crossing is off C; a line gives $m + 2k = 6$, $-2 + 4/3 + m/2 + 5k/6 = 4/3$, hence $m = -6$, off D. On D, $J = G_7$ gives $m + 2k = 5$, $-2 + 2/3 + 11/12 + m/2 + 5k/6 = 4/3$, hence $m = -4$.

Lemma 4.2. An irreducible factor of rank 4 occurs in some point stabilizer. The possible groups of rank 4 are $A_4, B_4, D_4, G(4, 2, 4), G(4, 4, 4), F_4, H_4, G_{29}, G_{31}, G_{32}$. Every smaller irreducible factor enters one of these groups as a parabolic subgroup, so every reflection has order 2 or 3.

Proof. Suppose first that no point stabilizer has an irreducible factor of rank 4. Follow an irreducible factor through successive absorptions; equal rank enlargements already lie in its full stabilizer. If the rank stops below 4, [15, Lemma 2.12, Proposition 2.14] exclude ranks $3, 1$ by collision geometry and give $Y = \mathbb{P}^2$, $N_{Y/X} = \mathcal{O}(-1)^2$ in rank 2. Equal degrees q occur only for $G(m, 2, 2)$, even $m \geq 4$, and G_7, G_{11}, G_{19}, with $q = m, 12, 24, 60$. At a point x of a persistent plane, its generic factor is a direct factor of the full ambient stabilizer, since an absorption would give an irreducible factor of rank at least 3. The effective local groups on the plane are therefore products of cyclic groups or the irreducible groups of rank 2 with equal basic degrees listed above. Their branch germs are nodes or ordinary triple points of types $[2, 2, m/2]$, $[2, 3, 3]$, $[2, 3, 4]$, and $[2, 3, 5]$, by [28, Table 1]. Several local branches may belong to the same irreducible curve. Write $\Delta_Y = \sum_i \beta_i D_i$, where $\beta_i = 1 - b_i^{-1}$, and let d_i, g_i, and δ_i be the degree, the genus of the normalization, and the sum of the delta invariants of D_i. In [28, equations (5), (7), and (8)], identifying r local branches of index b_i changes θ_x by $-3\beta_i\binom{r}{2}$. Here Γ_x is the effective local group on the plane, and ρ_1, ρ_2 count the mirrors above two of these branches. Their intersection multiplicity is one, so pulling back gives $b_i^2 \rho_1 \rho_2 = |\Gamma_x|$. Thus [28, Propositions 3.7 and 3.9] gives

$$0 = \mathcal{P}(Y, \Delta_Y) = \frac{3}{2} \sum_i \beta_i \big(d_i(d_i - 1) - 2\delta_i\big) = 3 \sum_i \beta_i (d_i - 1 + g_i).$$

Every component is consequently a line, so the hypotheses of [28, Theorem 1.1] hold. Its two arrangements have automorphic degrees $1/6$ and $1/2$. Since $q\lambda = 1$ and the generic factor is irreducible, $q = 6$ and the arrangement is the complete quadrilateral. A point of type $[2, 3, 3]$ supplies a persistent G_7 factor, whose fixed plane has degree $1/12$, contradicting the same conclusion. This proves the first assertion. Sections 4.1.1–4.1.2 exclude every other rank 4 group. For a smaller factor never entering rank 4, the maximal 2-adic reflection order and

Section 4.1 exclude G_7, G_{11}, G_{19} and leave $q = 2s$ with odd $s \geq 3$. An order divisible by 4 gives $3/q = \deg(K_Y + \Delta_Y)$ a larger denominator valuation than its branch coefficients. For index $b \geq 5$, branch degree d and ν nodes, proportionality and Bézout give

$$6d - 4 + 2b\lambda d \leq 2d^2 + 2b\lambda d - 4\nu \leq d(3 + 3\lambda - (1 - b^{-1})d), \quad \lambda = 1/(2s),$$

forcing $d = 1$, $b/s \in \mathbb{Z}_{>0}$, $b(2b-3) \leq 2s$, a contradiction. Thus $3d_2 + 4d_3 = 18 + 9/s$, $s = 3, 9$. Divisibility of each index 3 degree by 3 excludes $q = 18$ and $(q, d_2, d_3) = (6, 7, 0)$. For $(d_2, d_3) = (3, 3)$, a component of degree $1 \leq d \leq 3$, with c cusps, satisfies $2d^2 + d - 9p_a(d) \leq 2d^2 + d - 9c - 4\nu \leq 3d/2$; the lower bounds $3, 10, 12$ exceed $3/2, 3, 9/2$. □

Lemma 4.3. For every coincidence plane $Z_I \simeq \mathbb{P}^2$, $|I| = 3$, write $L|_{Z_I} = \lambda_I \mathcal{O}(1)$. Then $\lambda_I \geq 1/2$ and $2L - H$ is semiample. In particular, $a \leq 2t$ for every complete curve.

Proof. Lemma 3.1 gives $\lambda \geq 1$ for general ranks $0, 1$ and $q\lambda \geq 1$ for the least invariant Taylor degree q in rank 2, with regular maps also at larger stabilizers. For $\lambda < 1/2$, Section 4.1 leaves the following possibilities, with d the total degree of the upper locus:

general group	upper group	degree equation
$G(4,2,2)$	$G(4,2,3)$	$8\lambda = 2 + 3d/2, \quad d = 0, 1$
G_4	G_{25}	$10\lambda = 2$ if no upper locus
μ_3^2	G_{25}	$6\lambda = 2 + 5d/2, \quad d = 0$

.

For the first row, invariants $v + S, vS + P^2, u^2P$, $v = u^4$, $S = y_1^4 + y_2^4$, $P = (y_1y_2)^2$, give $d_2'/2 + 2d_3'/3 = (60 - 3d)/16$, impossible after multiplication by 6. For G_4, the upper axis $(v^2, v^3, 0)$ is nonimmersed; its absence contradicts $4\lambda \geq 1$. For μ_3^2, the upper differential $\left(\begin{smallmatrix} 0 & -v \\ 24v^2 & 0 \end{smallmatrix}\right)$ gives $\lambda = 1/3$ and $(d_2, d_3) = (8, 0), (4, 3), (0, 6)$. An index 3 component of degree $d \leq 6$ requires $9c + 4\nu = 2d(d+1)$ and $c + \nu \leq (d-1)(d-2)/2$. For $d \leq 5$ these are incompatible; for $d = 6$, the only pairs $(c, \nu) = (0, 21), (4, 12), (8, 3)$ still violate the bound. Hence $d_3 = 0$, but the normalized orders $1, 2, 4$ make each index 2 degree divisible by 3, contrary to $d_2 = 8$. For multidegrees (w_i) of $2L - H$, the triple degrees $d_I = 2\lambda_I - 1 \geq 0$ give $w_i = d_I + d_J \geq 0$ by complementary triples; [15, Proposition 2.6] proves semiampleness. □

4.2 The finite normal matrices

The following finite recipe uses coarse coordinates. Expand $F_j(uv_0 + y) = F_j(uv_0) + \sum_i c_{ji}(u) g_i(y)$ modulo $(g_1, \ldots, g_s)^2$, for ambient and normal basic invariants F_j, g_i. Adjoin the tangent column, remove its common factor, eliminate that column, and insert the character frames specified below. Cofactors then give inverse rows and their restrictions to successive kernels. Every remaining operation is holomorphic and invertible, so the conclusions hold under arbitrary analytic isomorphisms of quotient germs. Here are all inputs for the larger groups; the last triple gives the determinant order, highest pole and pole on the preceding kernel.

$T \subset G$	degrees of G	degrees of T	orders
$G_{25} \subset G_{32}$	$12, 18, 24, 30$	$6, 9, 12$	$(8, 3, —)$
$A_3 \subset A_4$	$2, 3, 4, 5$	$2, 3, 4$	$(3, 1, 1)$
$A_3 \subset D_4$	$2, 4, 4, 6$	$2, 3, 4$	$(3, 1, 1)$
$B_3 \subset B_4$	$2, 4, 6, 8$	$2, 4, 6$	$(3, 1, 1)$
$B_3 \subset F_4$	$2, 6, 8, 12$	$2, 4, 6$	$(7, 3, 2)$
$H_3 \subset H_4$	$2, 12, 20, 30$	$2, 6, 10$	$(22, 10, 7)$
$G(4,4,3) \subset G_{29}$	$4, 8, 12, 20$	$4, 3, 8$	$(7, 3, 3)$

Every next pole in the table is measured on the preceding kernel. For $G_{25} \subset G_{32}$ use $r_j z = z - (Cz)_j \varepsilon_j$, with

$$C = \begin{pmatrix} 1-\omega & \omega^2 & 0 & 0 \\ -\omega^2 & 1-\omega & -\omega^2 & 0 \\ 0 & \omega^2 & 1-\omega & \omega^2 \\ 0 & 0 & -\omega^2 & 1-\omega \end{pmatrix}, \qquad \omega^3 = 1, \quad \omega \neq 1.$$

The first three reflections fix $v_0 = ((2\omega+1)/3, 1, (4\omega+2)/3, 1)$ and act normally on $z_4 = 0$. Use the 240 forms in $e_4^* G_{32}$ and $(1,2,3)G_{25}$ for the respective power sums; separate the square of degree 6 in degree 12. Normalizing the first coefficient row to $(1,1,1,1)$ gives determinant $-19489164500000/1162261467$. The first axis values are $(56/9)v^2, (1456/243)v^3$, $v = u^6$, and the degree 9 frame is $u^3 \partial_{f_9}$. Thus $M(v) = \operatorname{diag}(v^2, v^3, v^4) A \operatorname{diag}(1, 1, v^{-1})$, $\det A \neq 0$, with a simple tangent zero.

For $A_3 \subset A_4$ use $(u, u, u, u, -4u)$ and $\sum y_i = 0$; the power sums have values $(20u^2, -60u^3, 260u^4, -1020u^5)$. For $A_3 \subset D_4$ use (u, u, u, u), $v = u^2$, and character frame $u\partial_{q_3}$. The matrices are

$$M_A = \begin{pmatrix} 15u/2 & 1 & 0 \\ -20u^2 & 4u & 1 \\ 275u^3/2 & 10u^2 & 5u \end{pmatrix}, \quad M_D = \begin{pmatrix} 4v & 4v & 1 \\ 0 & 4v/3 & 0 \\ 0 & 8v^2 & 12v \end{pmatrix}.$$

The determinants are $\det M_A = 625u^3/2$ and $\det M_D = 64v^3$. The row associated with the invariant of greatest degree and the next row restricted to its kernel remain nonzero after pole correction at every stalk; the last zero has order 1, and the A_4 tangent zero is simple. All three D_4 types are conjugate by the root preserving transformations $\frac{1}{2}\left(\begin{smallmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{smallmatrix}\right)$ and $\operatorname{diag}(-1, 1, 1, 1)$.

For $B_3 \subset F_4$ take the short roots $\{\pm\varepsilon_j\} \cup \{\frac{1}{2}\sum \epsilon_j \varepsilon_j : \epsilon_j = \pm 1\}$ and their power sums. On $u\varepsilon_1$, $v = u^2$, elimination gives rows $(-3v^2, 15v, 18)$, $(0, -15v^2, 45v)$, $(0, 0, -405v^3/8)$. For $u(\varepsilon_1 + \varepsilon_2)$ multiply the rows by $-8, 16, -64$; axis values $6v, 12v$ prove immersion, and the last zero order is 2. For $B_3 \subset B_4$ use $\left(\begin{smallmatrix} v & 1 & 0 \\ 0 & v & 1 \\ 0 & 0 & v \end{smallmatrix}\right)$, of last zero order 1. For $H_3 \subset H_4$ use the Gram matrix of diagonal 2 and successive off-diagonal entries $-\phi, -1, -1$, $\phi = (1+\sqrt{5})/2$, and axis $(1+\sqrt{5}/2, (3+\sqrt{5})/2, (5+\sqrt{5})/4, 1)$. The 120 ambient and 30 normal root sums give coefficients of exponent $(d-e)/2$ in $v = u^2$. The nonzero pivots are $-(21\sqrt{5}+63)v^5/512$, $(3740\sqrt{5}+8415)v^7/1024$, $(22297275\sqrt{5}+49864815)v^{10}/1048576$.

For $G(4,4,3) \subset G_{29}$ use $r_j z = z - (Bz)_j \varepsilon_j$, with

$$B = \begin{pmatrix} 2 & -1 & -i & 0 \\ -1 & 2 & i-1 & 0 \\ i & -i-1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

The first three reflections fix $((1+i)/2, 1-i/2, 3/2, 1)$; the 160 forms in $z_1 G_{29}$ give tangent and normal determinant $128957475465i/524288$. For $v = u^4$, elimination gives rows $(-81v/8, -9iv^2/4, -81/4)$, $(0, -1183iv^3/384, -3549v/16)$, $(0, 0, -4037365v^3/4096)$. The degree 3 character is $-i$, with frame $u^3\partial_P$, and last zero order 1. At $G(4,4,4)$ use $\left(\begin{smallmatrix} v & 0 & 1 \\ 0 & v & 0 \\ 0 & 0 & v \end{smallmatrix}\right)$. The unique parabolic class and nonzero first axis invariant proportional to v give a single smooth branch. Distinct basic degrees make the full normalizer preserve the flag.

4.3 The groups B_4 and G_{32}

Lemma 4.4. A B_4 axis contains an F_4 or G_{29} point. A $G(4,2,4)$ axis contains a G_{31} point, and a $G(4,4,4)$ axis contains a G_{29} point.

Proof. If a B_4 axis contains neither exceptional upper group, its only upper groups are B_4 and products. [15, Proposition 3.4] gives $2x + R/2 + \sigma = 2t$ and $a = -2x - 12t + 3R$. These equations imply $a \le -14x$, so $C = \mathbb{P}^1$. Put $k = 2t \in \mathbb{Z}_{>0}$ and $j = R - 2k$. Then $a = 2 + 3j$, $\sigma = 2 - j/2$, and the quotient associated with the invariant of greatest degree has degree $h = j - k$. Lemma 4.3 gives $k \ge 2 + 3j$, hence $j \ge 0$ and $h < 0$. The monomial calculation in Section 4.1.1 gives Hom dimension 1, attained by the surjective projection to the coordinate of greatest invariant degree. [15, Proposition 2.2] therefore gives two coincident labels, a common bundle of degree $-(k - j)$, and $a \ge 3(k - j)$. This contradicts $2 + 3j \le k \le 2j + 2/3$.

For $G(4, p, 4)$, $p = 2, 4$, assume the corresponding exceptional group is absent. The upper groups are then $G(4, p, 4)$ and products, and $k = 4t$ is a positive integer. The common equations give $g = 0$, $k \le 3$, and the three rows

$$(p, k, R, a, \sigma) = (2, 1, 3, 5, 1/4), (2, 2, 4, 3, 0), (4, 1, 3, 5, 1/4).$$

The first and third have $0 < \sigma < 1/2$; the second contradicts $a \le 2t$. □

Lemma 4.5. The group G_{32} does not occur, and every reflection has order 2.

Proof. Take the axis of G_{25} at a G_{32} point. Let R count G_{32} points and s_b count points with stabilizer $G_{25} \times \mu_b$, for $b = 2, 3$. The calculation in Section 4.2 gives effective order 6, characters $(1, -1, 1)$ on degrees $6, 9, 12$, a simple tangent zero, normal determinant order 8, and highest pole 3. Consequently, with $k = 6t \in \mathbb{Z}_{>0}$,

$$2t = 2x + 5R/6 + s_2/2 + 2s_3/3, \qquad a = -2x - 27t + 17R/2, \qquad h = -12t + 3R.$$

Eliminating t gives $a = -29x - 11R/4 - 27s_2/2 - 18s_3$, hence $g = 0$. The positive integer a and $a \le k/3$ leave exactly

$$(R, s_2, s_3, t, a, h) = (2, 2, 1, 2/3, 1, -2), (7, 1, 0, 13/6, 3, -5).$$

Both nonzero inverse maps have target degree less than $-a$, contrary to [15, Proposition 2.2]. Lemma 4.2 now excludes every reflection of order 3. □

4.4 The plane of degree 1/2

A compact plane quotient of degree 1/2 has branch indices 2 and reduced branch degree 9. For nodes, cusps $(2, 3)$, contacts 2, cusps $(2, 5)$, and triple points, let the counts be ν, c, t_0, q_0, r. Their local groups $\mu_2^2, A_2, B_2, I_2(5), G(4, 2, 2)$ have orders $4, 6, 8, 10, 16$, δ invariants $1, 1, 2, 2, 3$, Milnor numbers $1, 2, 3, 4, 4$, and normalized orders $(2, 2), (1), (2, 2), (1), (4, 4, 4)$. For k components, $\sum g_i = 27 + k - \delta_B$, $\delta_B = \nu + c + 2t_0 + 2q_0 + 3r \le 36$. Proportionality and $\chi_{\mathrm{orb}} = 3/4$ give

$$4\nu + 8c + 12t_0 + 16q_0 + 15r = 180, \qquad 180\nu + 320c + 450t_0 + 576q_0 + 585r = 7020.$$

Thus $c = 9c_1$, $q_0 = 5q_1$, $r = 4r_1$, and

$$t_0 = 12 - 4c_1 - 8q_1 - 4r_1, \quad \nu = 9 - 6c_1 + 4q_1 - 3r_1, \quad \delta_B = 33 - 5c_1 - 2q_1 + r_1.$$

The conditions $c_1 + 2q_1 + r_1 \le 3$, $\nu \ge 0$, $\delta_B \le 36$ leave exactly the nine rows below. A line or smooth conic of degree d satisfies $n_i + 2t_i + 2r_i = d(9 - d)$, $n_i/2 + t_i/2 + 3r_i/4 = d + 2$. Consequently lines have $(n_i, t_i, r_i) = (0, 0, 4), (2, 1, 2), (4, 2, 0)$ and conics need $t_i \ge 5$. For the union B^* of duals of nonlinear components and l lines, $D = \deg B^* = 72 - 2\nu - 3c - 4t_0 - 5q_0 - 6r$ and $\delta(B^*) = (D - 1)(D - 2)/2 + \delta_B - 28 - l$. A contact not involving a line, or a cusp $(2, 5)$, contributes at least 2 to $\delta(B^*)$, additively even at a common dual point. For a cusp write $x = u^2$, $y = a_4u^4 + a_5u^5 + \cdots$, $a_5 \ne 0$; its dual has type $(2, 5)$ after subtracting the quadratic term if $a_4 \ne 0$, and $(3, 5)$ otherwise.

row	(c_1, q_1, r_1)	exclusion or equality
1	$(0,0,3)$	$\delta_B = 36$: nine lines and twelve ordinary triple points.
2	$(1,0,1)$	No nodes, hence no line contact; dual upper/lower bounds $7, 8$.
3	$(0,0,2)$	$k \geq 8$: one conic and seven lines; five required contacts exceed four.
4	$(1,0,0)$	Too few nodes for a line; dual bounds $15, 16$.
5	$(0,0,1)$	$k = 7$: a cubic supplies fewer than eight contacts; the partition into two conics has impossible tangent distributions.
6	$(1,1,0)$	No contacts; dual bounds $8, 10$.
7	$(0,0,0)$	$k = 7$: at most six contacts; $k = 6$, partitions $(4,1^5), (3,2,1^4)$: at most eleven.
8	$(0,1,1)$	No contacts; dual bounds $5, 10$.
9	$(0,1,0)$	$k \geq 4$: five cusps $(2,5)$ force $(6,1,1,1)$; the three lines need six contacts rather than four.

For row 5, the conics need two mutual contacts, six line contacts, and $(t_i, n_i, r_i) = (5,0,2)$. Their four triple points are distinct; distributions of lines touching two, one, or no conic are $(1,4,0), (2,2,1), (3,0,2)$. A line touching one conic joins the other's two triple points, so there are at most two; a line touching neither joins both pairs, excluding the last two distributions as well. The remaining partition $(2,2,2,1,1,1)$ in row 7 requires two contacts per conic pair. Their dual intersections are exhausted by the common tangents; a line touching two conics would give three tangent branches, an excluded local type. Only row 1 remains.

Lemma 4.6. If $\lambda_I = 1/2$, the general group on Z_I is μ_2^2, and every stabilizer on the fixed plane is $\mu_2^2 \times J$. The induced branch divisor consists of nine lines with twelve ordinary triple points and no other singularities. Two disjoint triples cannot both have degree $1/2$. The integral semiample bundle $\mathcal{A} = 5(2L - H)$ has strictly positive multidegrees v_i. A curve fixed by an irreducible group of rank 3 avoids every such plane.

Proof. Lemma 3.1 excludes ranks $0, 1$, and $G(4,2,2)$ would require $8\lambda = 2 + 3d/2$, $d = 4/3$. For the remaining groups the quadratic Taylor map to $\mathcal{O}(-1)^2$ has numerically trivial coefficients, so its nonzero coefficient or determinant never vanishes. Absorption would lower its rank: nonreal factors have no quadratic invariant, and a real irreducible quadratic form cannot lie in the plane ideal. Thus the factors stay direct, and Lemma 4.2 leaves only μ_2^2. Section 4.4 gives the branch divisor. Disjoint triple planes meet transversely at one point and share no reflection, since otherwise they lie in a smooth threefold. Their full group would be μ_2^4, inducing an excluded node. Writing $q = \sum v_i/2$, $d_I = q - \sum_{i \in I} v_i$, a zero v_i gives complementary zero triples and this excluded pair; hence $v_i \geq 1$. Every factor of $\mu_2^2 \times J$ has rank at most 2, proving the final assertion. □

4.5 The group G_{31}

Lemma 4.7. Let $T = G(4,2,3)$ be the full stabilizer of a generic point of a fixed curve in a smooth finite reflection quotient of dimension 4. Assume that its full special stabilizer is $G(4,2,4)$, G_{31}, or $T \times A_1$. The regular frames corresponding to the basic polynomial invariants have degrees $4, 6, 8$. At the first two types their inertia characters are $(1, -1, 1)$, and the character correction of the middle coarse frame is $-1/2$. The following orders hold, independently

of arithmeticity or compactness:

	ord(df)	ord($\det M$)
$G(4,2,4)$	0	3
G_{31}	1	10
$T \times A_1$	0	0

	highest pole	middle pole on its kernel	lowest zero
$G(4,2,4)$	1	1	1
G_{31}	4	4	2
$T \times A_1$	0	0	0

The unrestricted middle inverse direction at G_{31} has pole order 5.

Proof. Take $f_4 = \sum y_j^4, f_6 = (y_1y_2y_3)^2, f_8 = \sum_{j<k} y_j^4 y_k^4$. For $G(4,2,4)$, use invariants $(v + f_4, vf_4 + f_8, u^2 f_6, vf_8 + f_6^2)$, $v = u^4$, and frame $(\partial_{f_4}, u^2\partial_{f_6}, \partial_{f_8})$; removing the tangent gives M_U below. For G_{31}, the full normalizer is $T \times \mu_4$ [10, Table 8 and proof of Lemma 5.13]. From the roots and coroots of its Table 2, generate $I - {}^t\mathsf{A}_j\mathsf{B}_j$, $1 \le j \le 5$, and take power sums $8, 12, 20, 24$ of the 240 dual root forms and $4, 6, 8$ of the 192 forms in $(1,2,3)T$, subtracting a multiple of f_4^2 in degree 8. The basis, coefficient matrix, and normal matrices are

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1-i & (1+i)/2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 21/2 & -33/2 & 33 & 231 \\ 10795/2 & -41021/14 & 9690/7 & 226746 \\ 105165 & 129789/5 & 4807 & -19530841/5 \end{pmatrix},$$

$$M_U = \begin{pmatrix} v & 0 & 1 \\ 0 & v & 0 \\ 0 & 0 & v \end{pmatrix}, \quad M_R = \operatorname{diag}(v^2, v^4, v^5) A \operatorname{diag}(1, 1, v^{-1}).$$

Here A is the lower Schur complement of C, with $A_{11} = -27$, leading minor $2070090/7$, determinant -2066330716560, and last cofactor column ${}^t(3625040628/7, -4491828, 2070090/7)$. The generator $u \mapsto iu$ relative to automorphy gives characters $(1, -1, 1)$; the G_{31} axis is (v^2, v^3, v^5, v^6) up to nonzero rescaling. For $B = A^{-1}$, the kernel of the projection to the coordinate of greatest invariant degree is $c_3 = -(B_{31}v^3c_1 + B_{32}vc_2)/B_{33}$, where the middle row becomes $v^{-2}(B_{21} - B_{23}B_{31}/B_{33})c_1 + v^{-4}(B_{22} - B_{23}B_{32}/B_{33})c_2$. Its last coefficient is nonzero by $A_{11} \neq 0$, giving poles $4, 4$, rather than the unrestricted middle pole 5; the saturated first column and axis derivative have orders $2, 1$, giving $T_C(D_R)$. Products give the identity matrix; clearing the displayed poles leaves primitive rows on every stalk, proving the table. □

Lemma 4.8. The group G_{31} does not occur.

Proof. On the $G(4,2,3)$ axis, let U, R, s count $G(4,2,4), G_{31}, T \times \mu_2$, with $R \ge 1$. Lemma 4.7 gives saturated tangent $T_C(D_R)$, normal determinant order 10, and highest pole 4; the next pole is 4 on its kernel, but 5 on the whole bundle. The everywhere surjective flag has $h_8 = -8t + U + 4R$, $h_6 = -6t + U/2 + 7R/2$, $h_4 = -4t + U + 2R$, with $2t = 2x + 3(U + R)/4 + s/2$ and $a + 2x - R = -18t + 5U/2 + 19R/2$. For $V = 2t - a \ge 0$, $11h_8 = -4V - 2U - R - 2s < 0$; put $e = -h_8$. Three coincident labels give a plane tangent quotient J of degree $3a + 2x - R \ge a$ by $R \le 2a + 2x$. Both successive maps kill J, since $\deg J - h_6 = U/2 - 3e/2 + 2a > 0$, and $e \le a$. With $b = a - e$, $\delta = e - R$, the degrees give $11\delta \ge 20b$, $x = -e/2 - \delta + b/2 < 0$, $b = e + 2\delta - 2$, $20e + 29\delta \le 40$. The sole row $(a, e, U, R, s) = (2, 2, 6, 2, 0)$ is a smooth conic with two tangent zeros, impossible. Thus two labels coincide, the quotient associated with the invariant of greatest degree is M, and [15, Proposition 2.3] gives $h_6 \ge e$, equivalently

$$\begin{aligned} t &= (e + U + 4R)/8, & a &= (6R + U - 9e)/4 - 2x, \\ s &= (e - 2U + R)/2 - 4x, & 2R &\ge U + 7e. \end{aligned} \tag{4.1}$$

The collision divisor is the blow-up of $\mathbb{P}^3$ at five points. For its blow-down $\gamma : C \to \mathbb{P}^3$, put $d = \deg \gamma^* \mathcal{O}(1)$, $B_0 = \sum \beta_j$ for the five brackets; then $e = d - B_0$, $a = 2d + e$. The image is not a line, by $R \geq 1$ tangent zeros, so a general plane projection has positive dual degree at most $m = 2d + 2x - R$. Since $4m = 6x - 5V - s$, $m, x > 0$. For $\mathcal{A}$ of Lemma 4.6, put $q = \sum v_i/2$, $\rho = q - v_1 - v_2$; then $1 \leq v_j \leq \rho$ for $j \geq 3$, $5V = e\rho + \sum_{j=3}^{7} v_j \beta_j$, and $v_j = \rho$ implies $\beta_j = 0$. Every outer triple of weight sum q gives a cross-ratio $[1j][kl] : [1k][jl]$ with no base point on C, by avoidance of the planes of degree $1/2$. Its degree is $e + \beta_j + \beta_k + \beta_l > 0$, and all G_{31} points ramify, giving $R \leq 2(e + \beta_j + \beta_k + \beta_l) + 2x$. Four zero outer brackets give $R \leq 3e + 3x$ by three cross-ratios with disjoint base loci; fewer than three nonconstant ones force rational normalization. An identical outer pair gives $R \leq 2e + 2x$, and two identical pairs contradict a positive constant cross-ratio degree. The exclusions before the finite test are

condition	inequalities	contradiction
$R \leq 3e + 3x$	$16m + 20e \leq 14x, \quad 4m + 15e \geq 16x$	$32m \leq -22x$
$\rho \geq 5$	$53m + 65e \leq 62x$	$107m \leq -22x$
$\rho \geq 3,\ v_j \geq 2$	$33m + 30e \leq 32x$	$25m \leq 0$

The first row excludes $a = 3e$ and $\rho = 1$. The eleven remaining multidegrees are parametrized below; exponents mean repetition and the first pair is ordered.

ρ	j	$(v_3, \ldots, v_7)$	(v_1, v_2)
2	$1, 3, 4$	$(1^{5-j}, 2^j)$	$(\lfloor (1+j)/2 \rfloor, \lceil (1+j)/2 \rceil)$
$3, 4$	$0, 1, 3, 4$	$(1, (\rho-1)^{4-j}, \rho^j)$	$(\lfloor (2\rho - 3 + j)/2 \rfloor, \lceil (2\rho - 3 + j)/2 \rceil)$

This follows from $v_1 + v_2 = \sum_{j=3}^{7} v_j - 2\rho$ and every triple inequality; the two all-2 outer rows at $\rho = 2$ have $B_0 = 0$ and are already excluded. For $\rho = 2, j = 1$, six cross-ratios give $R \leq 2e + B_0 + 2x$. Set $S = 12s$, $Q = 6(2e + B_0 + 2x - R)$, and $E = 21e + 5U + 80x - 52B_0 = 0$. Substitution of (4.1) and $a = 3e + 2B_0$ gives $828e + 7S + 106Q - 3E = -96x$. For $j = 3, 4$ use $R \leq 2e + 2x$, $R \leq 3e + 3x$. For $\rho = 3, 4$, $j = 0, 1, 3$, let l be the bracket degree for weight 1 and H_0 their sum for weight $\rho - 1$. Averaging gives $R \leq 2e + 3H_0/2 + 2x$. Put $Q = 6(2e + 3H_0/2 + 2x - R)$ and $E_\rho = (45 - 12\rho)e + 5U + 80x - 52l - (28 + 12\rho)H_0 = 0$. Substitution of (4.1) and $a = 3e + 2(l + H_0)$ gives

$$(180\rho - 150)U + 16\rho S + (8\rho + 115)6m + (20\rho - 40)Q + 15E_\rho = -(528\rho - 1500)x.$$

The left side is nonnegative and the right side negative; $j = 4$ has four zero outer brackets and uses the first exclusion. □

4.5.1 Determinant valuations along an A_3 curve

Suppose the last A_3 kernel, corresponding to the invariant of degree 2, is the tangent quotient of a coincidence plane Z. Let $\eta : S \to Z$ resolve an inverse image component of degree e, with $L|_Z = \lambda \mathcal{O}_Z(1)$. For S_4 invariants on $\sum y_i = 0$, the plane equations start as $F_3 = q_3 + O(q^2)$, $F_4 = q_4 + c(u)q_2^2 + O_{>4}(y)$. Their differential wedge has degree at least 5 for general rank 0. For rank 1 write $y = (v + r, v - r, z - v, -z - v)$; then $q_2 = 4v^2 + 2r^2 + 2z^2$, $q_3 = 6v(r^2 - z^2)$, $q_4 = 4v^4 + 12v^2(r^2 + z^2) + 2r^4 + 2z^4$. The r^2 coefficients in F_3, F_4 have degrees at least $1, 2$, so their determinant with the tangential differential has degree at least 4. At a divisor with minimum normal valuation k, the leading vector satisfies $q_3(y^0) = q_4(y^0) + c(u)q_2(y^0)^2 = 0$ and $q_2(y^0) \neq 0$, since otherwise all invariants vanish. Thus $\eta^* C$ has valuation $2k$, and the two determinants have valuations at least $5k, 4k$. Their saturated kernels are direct summands over the discrete valuation ring, including a critical divisor after removal of the differential's common factor. Exceptional curves have zero intersection with a general pulled back line; hence the determinant divisors contain $5\eta^* C/2$, $2\eta^* C$. Gauss bounds their degrees by $(2\lambda - 2)e$, $(3\lambda - 2)e$, as used in Lemma 4.10.

4.6 The stronger degree bound and the groups A_4, D_4

Lemma 4.9. Every $\lambda_I \geq 1$, and $L - H$ is semiample.

Proof. Lemmas 4.4, 4.8 exclude $G(4,2,4)$. The remaining seven groups lack the $G(4,2,2)$ factor required at the twelve triple points of a plane of degree $1/2$ by Lemma 4.6. All reflection orders are 2, so $10L = B_X - 2H$ is integral. In general rank 2, the induced indices $2, 4$ give $3\lambda = -3 + d_2/2 + 3d_4/4$, hence $12\lambda, 10\lambda \in \mathbb{Z}$ and $2\lambda \in \mathbb{Z}$. The excluded equality and Lemma 3.1 for ranks $0, 1$ yield $\lambda \geq 1$. For multidegrees v_i of $5(2L - H)$, $d_I = 10\lambda_I - 5 \geq 5$ and $v_i = d_I + d_J \geq 10$; the multidegrees $v_i - 10$ and triple degrees $d_I - 5$ prove semiampleness of $10(L - H)$ by [15, Proposition 2.6]. □

Lemma 4.10. At least one point has stabilizer F_4, H_4, or G_{29}.

Proof. Otherwise Lemma 4.4 leaves A_4, D_4. On an A_3 curve count $A_4, D_4, A_3 \times \mu_2$ by R_A, R_D, s. The matrices of Section 4.2 give tangent $T_C(D_A)$ and a surjective quotient Q of rank 2, with $h_3 = -3t + R_A + R_D/2$, $h_4 = -4t + R_A + R_D$. The degree calculation gives $2t = 2x + (R_D + s)/2$, $a = -2x - 9t + 4R_A + 5R_D/2$, $h_3 + h_4 = a/2 - 3t/2 - s/4 < 0$. By [15, Proposition 2.4], $Q = M^2$, $\deg M = -a$, with plane tangent kernel; hence $10a = 6t + s$, $a + 6R_A = 18x + 4s$, positive even a and $t/a \leq 5/3$. If $R_A \geq 1$, plane rank 2 is excluded by the effective degrees $1, 2$ of $A_2, 2A_1 \subset A_4$. For ranks $0, 1$, Section 4.5.1 and (3.1) give $5a/2 \leq 2\lambda - 2$ or $2a \leq 3\lambda - 2$, contrary to $a \geq 2$, $\lambda = t/a \leq 5/3$. If $R_A = 0$, the surjective degree h_3 map forces $h_3 \geq -a$, giving $12x + 5s/2 \geq 0$, $14x + 3s \leq 0$. Thus $x = s = 0$, $a = t = R_D/4$, but $\deg \det Q = -a$, contrary to $Q = M^2$. □

4.7 The arithmetic volume

Lemma 4.11. For a cocompact arithmetic lattice generated by reflections of order 2, put $d_F = [F : \mathbb{Q}]$, $\Delta = D_K / D_F^2$, $w = |\mu_5(K)|$, and $C_0 = 288/(44/7)^{14}$. Then

$$\frac{\chi_{\mathrm{orb}}}{5w\rho_{K/F}} \in \mathbb{Z}_{>0}, \qquad \rho_{K/F} = 2^{-4d_F}\zeta_F(-1)L_{K/F}(-2)\zeta_F(-3)L_{K/F}(-4),$$

and $\chi_{\mathrm{orb}} > 5wD_F^{12}\Delta^7 C_0^{d_F}$.

Proof. For $p : \mathrm{SU}(1,4) \to \mathrm{PU}(1,4)$, choose a maximal arithmetic group above $p^{-1}\Gamma$, with principal subgroup Λ. Its quotient is a 5-group [20, Sections 2.4 and 2.9]; every normalized order 2 reflection has $-r \in \Lambda$, so $\Gamma \subset p(\Lambda)$. Prasad's formula gives $\chi_{\mathrm{orb}} = 5wj\rho_{K/F}\prod_{v<\infty} e'(P_v)$, with $j = [p(\Lambda) : \Gamma]$ and $e'(P_v) \in \mathbb{Z}_{>0}$ [20, Sections 2.3 and 2.6–2.9, equations (7)–(8)]. The functional equations give $\rho_{K/F} = D_F^{12}\Delta^7(288/(2\pi)^{14})^{d_F}$ times an Euler product greater than 1: inert paired denominators $(1 - q^{-2})(1 + q^{-3})$, $(1 - q^{-4})(1 + q^{-5})$ are less than 1, as are the split and ramified denominators. Use $2\pi < 44/7$. □

Lemma 4.12. Under the hypotheses of Lemma 4.11, $\chi_{\mathrm{orb}} \neq 770$.

Proof. Section 4.7.1 proves this by the complete field filters and the nonintegral special values. □

4.7.1 The value 770

Suppose $\chi_{\mathrm{orb}} = 770$ in Lemma 4.11. For $F = \mathbb{Q}$, Meyer's theorem makes the indefinite rational form in ten variables isotropic, contradicting compactness [22, Chapter IV, Section 3.2, Corollary 2]. For $d_F \geq 5$, use $D_F^{1/d_F} > 34/5$ [20, Proposition 2(d)], $(34/5)^{12}C_0 > 1$, and $5((34/5)^{12}C_0)^5 > 770$. For $d_F = 2, 3, 4$, the discriminant bounds are $44, 230, 1226$. The following filters use the proved exhaustive small discriminant tables [20, Section 3.4, proof of

Theorem 1(iv)(1)–(3)]. For $d_F = 4$, $D_F = 725, 1125$ and $\Delta \le 2, \Delta = 1$ leave only $\mathbb{Q}(\zeta_{15})$, excluded by $25\rho_{K/F} = 45814652/30375 > 770$. For $d_F = 3, 2$, substituting $\Delta_{\max} + 1$ into the strict volume bound gives

D_F	49	81	148	169	229
$\Delta_{\max}$	14	5	2	1	1

,

D_F	5	8	12	13	17	21	24	28	29	33	37	40	41
$\Delta_{\max}$	40	17	8	7	4	3	2	2	1	1	1	1	1

.

The cubic candidates are $\mathbb{Q}(\zeta_7), \mathbb{Q}(\zeta_9)$. In degree 2, the tables exclude $13, 17, 29, 37, 41$ and leave, at $21, 24, 33$, $\mathbb{Q}(\sqrt{-3}, \sqrt{-7})$, $\mathbb{Q}(\sqrt{6}, \sqrt{-3})$, $\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. At 40, the only totally complex quartic of absolute discriminant 1600 is $\mathbb{Q}(\sqrt{5}, \sqrt{-2})$, with real discriminant 5.

For $D_F = 5, 8, 12, 28$, Minkowski gives real class number 1; at 28, the only ideal of norm 2 is generated by $3 + \sqrt{7}$. Write $K = F(\sqrt{\alpha})$ with totally negative α, whose square class is a product of units and principal prime generators. An odd valuation away from 2 contributes its prime norm to Δ, and above 2 contributes exponent $2v_{\mathfrak{p}}(2) + 1$. The permitted odd prime norms are $9, 5, 11, 19, 29, 31$ at $D_F = 5$ and $9, 7, 17$ at 8, so at most one occurs. At 12 the prime above 3 has mixed signs; at 28 no odd prime is allowed, and odd valuations above 2 exceed each bound. Signs at both embeddings leave the unique totally negative unit multiple up to squares and conjugation in each class:

F	α	Δ
$\mathbb{Q}(\sqrt{5})$	$-1, -3, -2 - \phi$	$16, 9, 5$
$\mathbb{Q}(\sqrt{2})$	$-1, -3, -5 - 2\sqrt{2}$	$4, 9, 17$
$\mathbb{Q}(\sqrt{3})$	-1	1
$\mathbb{Q}(\sqrt{7})$	-1	1

.

Here $\phi = (1 + \sqrt{5})/2$; the square residues modulo 4 in $\mathbb{Z}[\phi]$ have pairs $(1, 0), (1, 1), (2, 3)$, giving the displayed relative discriminants. For $\alpha = -5 - 2\sqrt{2}$, $\beta = 1 + 3\sqrt{2}$ gives $\beta^2 - \alpha = 8(3 + \sqrt{2})$, of valuation 6 above 2, versus 2 for 2β. Hensel makes α locally square; the split Euler factor gives $\chi_{\mathrm{orb}} > 5 \cdot 8^{12} \cdot 17^7 C_0^2 (16384/9765) > 868$. The complete twelve remaining fields are the nine biquadratic fields for

$$(u, v) = (5, -3), (5, -1), (2, -1), (2, -3), (3, -1), (-3, -7), (6, -3), (7, -1), (-3, -11),$$

and $\mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_7), \mathbb{Q}(\zeta_9)$. Only $\mathbb{Q}(\zeta_5)$ has $w = 5$, and every other field has $w = 1$. Use the exact Bernoulli sum $L(1 - j, \psi) = -(f_\psi^{j-1}/j) \sum_{u=1}^{f_\psi} \psi(u) B_j(u/f_\psi)$ for primitive characters, taking the three quadratic characters or the primitive even and odd cyclotomic characters. In the listed order the reduced denominators of $770/(5w\rho_{K/F})$ are $31, 587, 19, 18331, 23, 2, 551, 113, 3995, 293, 130429, 18628987$. None is 1, contrary to Lemma 4.11.

4.8 The last three groups

Lemma 4.13. The group H_4 does not occur.

Proof. Count $H_4, H_3 \times \mu_2$ on an H_3 axis by R, s. Section 4.2 gives $2t = 2x + (R + s)/2$, $a = -2x - 18t + 22R$, and $h_6 = -6t + 7R$, $h_{10} = -10t + 10R < 0$ by $44t \ge 45R + s$. For three labels, $3(\deg J - h_6) = 8a + 4x + R > 0$ makes both quotients kill the plane tangent. Their determinant and map associated with the invariant of greatest degree give $26a = 23R - 16x$, $2x - 11R \ge 0$, a contradiction. For two labels, [15, Proposition 2.3] gives $h_6 \ge e = -h_{10}$; integrality gives $e = 5b$, $b \in \mathbb{Z}_{>0}$. The complete degree conditions are $t = R + b/2$, $a = 4R - 2x - 9b$, $s = 3R - 4x + 2b$, $R \ge 8b$, but $a \le t$ gives $R \le 7b$. □

Lemma 4.14. Let $f : C \to X$ be an immersion from a curve of positive genus. Suppose a semiample GIT class B has nonnegative multidegrees satisfying every triple inequality and f^*B has degree zero. If the normal bundle has no nonzero map to a line bundle of negative degree, then $B \equiv 0$.

Proof. For nonzero (b_i), the triple inequalities give $b_i < q = \sum b_i/2$ and at least six positive weights; if one weight vanishes the others are equal. Two or three negative coincident sections give a line quotient of negative degree, and one gives $\deg f^*B = e(q - b_i) + \sum_{j\neq i} b_j\beta_j > 0$. Hence all relative tangent degrees are nonnegative and at least six vanish. Three sections are distinct, since a triple coincidence and nonvanishing remaining brackets imply $a = 0$. These sections trivialize the marked projective line bundle; its only possibly moving label ramifies in positive genus, contrary to immersion. □

Lemma 4.15. If an F_4 point exists and no G_{29} point exists, then $L \equiv H$.

Proof. Count $B_4, F_4, B_3 \times \mu_2$ on a B_3 axis by R_B, R_F, s, and put $q_0 = R_F - s$. Section 4.2 and $a \leq t$ give $a = -14x + 4q_0 + s$, $h_4 = q_0 - 4x$, $2h_6 = 3q_0 - R_B - 12x$, $R_B \geq 15q_0 + 2s - 60x$. For $h_6 = -e < 0$, two labels require $h_4 \geq e$ by [15, Proposition 2.3], giving $5e + s \leq 0$. Three labels have $\deg J - h_4 = (8a + 4x + R_F)/3 > 0$; the rank 2 quotient then gives $3R_B + 5R_F = 17a + 10x$, $R_F \geq 4a + 2x$, hence $-3R_B \geq 3a$. For $h_6 \geq 0$ the only solution is $q_0 = R_F = 4x$, $s = R_B = 0$, $a = t = 2x$, $(h_2, h_4, h_6) = (4x, 0, 0)$. Now $g \geq 2$ and the flag has no line quotient of negative degree; apply Lemma 4.14 to $L - H$. □

Lemma 4.16. The group G_{29} does not occur.

Proof. On the $G(4, 4, 3)$ axis let U, R, s count $G(4, 4, 4), G_{29}, T \times \mu_2$, with $R \geq 1$. Section 4.2 gives a single immersed branch and successive surjections, with $h_3 = -3t + U/4 + 9R/4$. The degree 3 character correction $-3(U + R)/4$ is included; the degree equations are $t = x + 3(U + R)/8 + s/4$, $a = -17x - 27U/8 + 5R/8 - 15s/4$, $h_8 = -8x - 2U - 2s$. The inequality $a \leq t$ forces $h_8 < 0$: the contrary for $x = 0$ gives $U = s = 0$, $a > t$; for $x = -1$, it gives $U + s \leq 4$, $a - t \geq 3 - s/4 + R/4 > 0$. Put $e = -h_8 = 8x + 2U + 2s \in 2\mathbb{Z}_{>0}$. For three labels the image is a smooth plane curve, $x = (a - 1)(a - 2)/2 - 1$, and $e \leq a$. The genus bound $8x - a = a(4a - 13) > 0$ excludes $a \geq 4$; integrality leaves only $a = 2$ among $a \leq 3$. For $a = 2$, the sole rows $(U, R, s, t) = (0, 6, 5, 5/2), (5, 3, 0, 2)$ contradict respectively $10(t/a) \in \mathbb{Z}$ and the next surjection, since $\deg J = 4 > h_3 = 2$ makes that surjection factor through degree -2. Thus exactly two labels coincide and $a \geq 3e$. For the nonzero class $10(L - H)$, nonzero by Lemma 4.12, let $q = \sum w_i/2$, $\rho = q - w_1 - w_2$, $\delta = t - a$. The triple inequalities give $\rho \geq 1$, $0 \leq w_j \leq \rho$ for $j \geq 3$, and

$$10\delta = e\rho + \sum_{j=3}^{7} w_j\beta_j, \qquad \delta \leq 8x/5 + s/10, \qquad \delta/e \leq 1/5.$$

Hence $1 \leq \rho \leq 2$, $w_1 + w_2 \leq 6$, $q \leq 8$, $w_i \leq 6$, $d_I \leq 8$. The multidegrees and triple degrees of $10(2H - L)$ are at least $14, 2$, so $2H - L$ is nef and $\chi_{\mathrm{orb}} = 5L^4 \leq 5(2H)^4 = 12320$. The scalar order $c \mid 10$ cannot be $1, 2$, since integrality of cL would force $5 \mid \rho$. Thus $c = 5, 10$, $\zeta_5 \in K$; the normalized central G_{29} element has trace $1 + 4i$, giving $\mathbb{Q}(\zeta_{20}) \subset K$ and $D_F \geq 2000^{[F:\mathbb{Q}]/4}$. Lemma 4.11 gives $\chi_{\mathrm{orb}} > 25(2000^3 C_0)^{[F:\mathbb{Q}]} \geq 25(2000^3 C_0)^4 > 12320$, since $2000^3 C_0 = 1192188995242382812 5/777727659178477568 > 1$. □

Theorem 4.17. No cocompact arithmetic lattice $\Gamma \subset \mathrm{PU}(1, 4)$ satisfies $X_7 \simeq \mathbb{B}^4/\Gamma$.

Proof. Lemmas 4.10, 4.13, and 4.16 force an F_4 point and exclude G_{29}. Lemma 4.15 gives $L \equiv H$. Since $H^4 = 154$, proportionality gives $\chi_{\mathrm{orb}} = 5H^4 = 770$, contrary to Lemma 4.12. □

5 Nonuniform realizations outside the Gaussian field

We retain the notation of Section 3.

Proposition 5.1. If X is the Baily–Borel compactification of a nonuniform arithmetic ball quotient, then its field of definition is $\mathbb{Q}(i)$.

We apply Lemmas 3.2 and 3.3 to the 35 coincidence planes.

5.1 The Coxeter groups

Lemma 5.2. Suppose that the full horizontal groups are A_3, B_3, the irreducible interior groups are among A_4, B_4, D_4, F_4, and the groups fixing a ball of dimension 2 are A_1^2, A_2, B_2. Suppose also that $2\mathcal{L}$ descends to X, $\mathcal{L} - H$ is nef, and the finite characters on the directions of indecomposable invariants have trivial inertia except for the usual real reflection characters. Then X has no cusps.

Proof. For a coincidence plane Z of generic rank 2, put $\mathcal{L}|_Z = \lambda\mathcal{O}_Z(1)$, $\lambda \geq 1$, $2\lambda \in \mathbb{Z}$. All inclusions use actual full parabolic stabilizers. The local determinants below give $10\lambda = 4 + 5\deg D_B$ for A_2, and $6\lambda - 2 = 2\deg D_B$, $\deg D_B \leq 2\lambda - 1$ for B_2, both impossible. For A_1^2, with loci D_A, D_B, D_P of $A_3, B_3, A_1 \times B_2$, they give

$$4\lambda - 2 = \tfrac{3}{2}\deg D_A + 3\deg D_B + \deg D_P. \tag{5.1}$$

For a B_3 curve, count $B_4, F_4, B_3 \times A_1$ by R_B, R_F, p, and cusps $1111, 1112, 1122, 1222$ by s_i. Put $x = g - 1$, $s = \sum s_i$, $(\alpha, \beta, \gamma) = (s_1 + s_2 + s_3 + 2s_4, s_1 + s_2 + 2s_3 + 2s_4, s_1 + 2s_2 + 2s_3 + 2s_4)$. The inverse differential gives everywhere surjective successive quotients of $K_2 \subset K_{24} \subset \mathcal{N}$, of ranks 1,2, with

$$(h_2, h_4, h_6) = (-2t + R_B + 2R_F + \alpha, -4t + R_B + 2R_F + \beta, -6t + R_B + 3R_F + \gamma). \tag{5.2}$$

The full characters and widths give

$$2t = 2x + s + (R_B + R_F + p)/2, \quad a = -2x - 12t + 3R_B + 7R_F + \alpha + \beta + \gamma. \tag{5.3}$$

The elimination below leaves, with cusps, only

$$(a, t, p, R_B, R_F; s_1, s_2, s_3, s_4; h_4, h_6) = (1, 3/2, 2, 0, 0; 0, 3, 1, 0; -1, -1). \tag{5.4}$$

Its plane has generic group A_1^2, $\lambda = 3/2$, $(\deg D_A, \deg D_B, \deg D_P) = (0, 1, 1)$. Without cusps $R_B = p = 0$, also excluding B_4 through its B_3 parabolics.

At a B_3 cusp use $x_1^2 = U$, $x_2^2 = T$, $x_3^2 = T - U$, $x_4^2 = T - cU$, $c \neq 0, 1$. Let S be the set of squared labels. A short reflection is actual exactly at a label in S; the other lift has nontrivial fiber character. All A_1^2 fixed components lie in the same plane [15, Lemma 2.17]. Two long reflections give an excluded A_3 curve, and two short reflections give two P branches; hence their lengths differ. The restricted rings for $1112, 1122, 1222, 1111$ are $\mathbb{C}[u, v]$, $\mathbb{C}[u, v]$, $\mathbb{C}[u, uv, v^2]$, $\mathbb{C}[u^2, uv, v^2]$. Embedding dimension leaves the three 1112 and one 1122 ends in (5.4), with horizontal order 2, boundary degree 2, width 1. Let D_Q be the A_1^3 locus. The normalizer test below excludes nontransverse interior D_P, D_Q intersections: their only possible ambient group $B_3 \times A_1$ gives another coincidence plane intersecting Z in a curve. The cusp branch counts (B, P, Q) are $(1, 0, 3)$ at 1112 and $(1, 1, 2)$ at 1122, with distinct directions $U, V, V - U, V - cU$. Thus $K_Z + \frac{1}{2}(D_A + D_B + D_P + D_Q) = 3\mathcal{L}|_Z$ gives $\deg D_Q = 13$. The cusp of D_P contributes multiplicity 2, leaving 11 transverse elliptic points of order 2, contrary to $3 = 2\mathcal{L} \cdot D_P \geq -2 + 1 + 11/2 = 9/2$. The actual $2\mathcal{L}$ has trivial cusp character. The curvature current of its invariant metric has zero Lelong number by the logarithmic cusp potential in [14, Section 2.2, equations (1)–(8)].

For A_3, count $A_4, D_4, A_3 \times A_1$ by R_A, R_D, p; all cusps are 1111. Saturate the tangent bundle to $T_C(D_A)$ at A_4. The normal corrections are $(h_4, h_3, h_2) = (-4t + R_A + R_D + s, -3t + R_A + R_D/2 + s, -2t + R_A + R_D + s)$. The comparison gives $2t = 2x + s + (R_D + p)/2$, $a = -2x - 9t + 4R_A + 5R_D/2 + 3s$. The last two directions give a surjective quotient Q, $\deg \det Q = a/2 - 3t/2 - p/4 < 0$. By [15, Proposition 2.4], $Q = M^{\oplus 2}$ on a three point coincidence plane, $\deg M = -a$. Hence $10a = 6t + p$, $a + 6R_A = 18x + 4p + 3s$, $1 \le \lambda \le 5/3$. Section 5.1.2 excludes rank 0 by $5a/2 \le 2\lambda - 2$; rank 1 forces $a = 1$, $x = -1$, $\lambda = 3/2$, then $R_D = -4R_A - 1$. The plane is therefore A_1^2. At an A_3 cusp, the polarized $\ker(E^4 \to E)$ has quotient $|4[0]| \simeq \mathbb{P}^3$ [21]. The components $(z, z, u - z, u - z)$, $u \in E[2]$, map through $|[0] + [u]|$ to noncollinear Veronese conics, incompatible with the boundary line $\mathbb{P}(T_p Z)$ and exceptional bundle $\mathcal{O}(-1)$ of a smooth cusp.

5.1.1 Coxeter groups and the local normal directions

For $A_2 \subset B_3$, expand the power sums at $x_i = u + y_i$, $\sum y_i = 0$, with $q_k = \sum y_i^k$, $v = u^2$ and coarse frame ue_3. Removing the tangent direction gives $\left(\begin{smallmatrix} 4v & 4v \\ 12v^2 & 20v^2 \end{smallmatrix}\right)$, of determinant $32v^3$. The A_3 axis $(12u^2, -24u^3, 84u^4)$ is not immersed. For B_2 the matrix is $\left(\begin{smallmatrix} v & 1 \\ 0 & v \end{smallmatrix}\right)$. For A_1^2 the determinant character contributes $-\deg D_A/2$, and local orders at $A_3, B_3, A_1 \times B_2, A_1^3$ are $2, 3, 1, 0$, proving (5.1) without splitting.

The companion matrix at B_4 and short root power sums at both F_4 axes give corrections $(1, 1, 1)$ and $(2, 2, 3)$ with nonzero minors (Section 4.2). The first three even Taylor coefficients of the cusp branch section products have a Vandermonde determinant. Thus the corrections in directions $(2, 4, 6)$ at $1111, 1112, 1122, 1222$ are $(1, 1, 1), (1, 1, 2), (1, 2, 2), (2, 2, 2)$, with width 1 at every common fixed point of trivial fiber character. Removing the raw 2222 fiber kernel gives 1111, replaces q by q^2, and halves the period. For $A_3 \subset A_4, D_4$, Section 4.2 gives $\det M_A = 625u^3/2$, $\det M_D = 64v^3$. The last two rows of M_D^{-1} in coarse frame ue_3 are $(0, 3/(4v), 0)$, $(0, -1/2, 1/(12v))$. These give independent extended coefficients of pole 1 for the highest and kernel directions; triality, tangent saturation at A_4, and the 1111 theta coefficients complete the surjectivity and character calculation.

For the full plane normalizers, enumerate orthogonal root pairs of pointwise stabilizer A_1^2 in A_4, B_4, D_4, F_4 and all rank 4 products of A_1, A_2, B_2, A_3, B_3. Restrict the other roots to the fixed plane and take invariants of $N_F(A_1^2)/A_1^2$ [10, Theorems 2.1, 3.9, 3.12, Table 5]. Roots orthogonal to the fixed space generate every stabilizer, proving exhaustiveness. Without A_3 branches or B_4 points, only $B_3 \times A_1$ gives a nontransverse P, Q pair; in F_4, equal root lengths give B_2, so A_1^2 uses both lengths.

5.1.2 Weighted vanishing on a surface

Let $\Sigma \subset X$ be a smooth surface containing C such that the projections of the inverse differential to the two coordinates of greatest invariant degree vanish along Σ. On a general line ℓ, put $\lambda = \mathcal{L} \cdot \ell$, $c = \det N_{\Sigma/X} \cdot \ell$, $d = C \cdot \ell$. Retain the critical divisor R in the Gauss inequality $K_\Sigma - R \ge 3\mathcal{L}|_\Sigma$ and in the conormal determinant classes $5\mathcal{L} - K_\Sigma + R$, $4\mathcal{L} - K_\Sigma + R$ for generic ranks 0,1. The rank 1 invariant normal class is $2\mathcal{L}$. For transverse order k, the orders of the curve equation and highest determinant are at most/at least $(3k, 6k), (2k, 7k)$ for B_3 and $(2k, 5k), (2k, 4k)$ for A_3. For B_3 the f_2^3 coefficient vanishes on the retained mirror. Thus

	rank = 0	rank = 1
B_3	$2d \le 2\lambda + c$	$7d/2 \le 3\lambda + c$
A_3	$5d/2 \le 2\lambda + c$	$2d \le 3\lambda + c$.

Retaining R prevents saturation from removing the zeros, and the general line avoids all cusp and blowup centers.

5.1.3 The equality cases for a B_3 curve

Put $q_0 = R_F - p$, $D_0 = 3s_1 + 2s_2 + s_3$, $U = s_1 + s_2$, $V = 2s_1 + s_2 + s_3 + s_4$. The comparison gives $a = -14x + 4q_0 + p - D_0$, $t = x + (R_B + q_0 + 2p + 2s)/4$, $h_4 = q_0 - 4x - U$, $h_6 = (3q_0 - R_B - 12x)/2 - V$. For $h_6 \geq 0$, $a \leq t$ forces $R_B = p = s_1 = s_3 = s_4 = 0$. Writing $b = s_2/3$ leaves $a = t = 2x + 2b$, $R_F = 4x + 2b$, $(h_2, h_4, h_6) = (4x + 3b, -b, 0)$. For $b > 0$, [15, Proposition 2.4] gives $b = 2a$, $x = -3a/2 < -1$; without cusps it gives $R_B = p = 0$.

For $h_6 < 0$ and two coincident labels of relative tangent degree $-e$, the bracket description gives $h_4 \geq e$ [15, Proposition 2.2]. The equality $R_B = -3(h_4 - e) - 2(t - a) - 2s_1 - s_2 - p - 2s_4$ forces $t = a$, $h_4 = e$, $s_3 = 5e/2$, $R_F = 4(e + v)$, $a = 3e + 2v$, $g = 1 + 3e/4 + v$, $4 \mid e$. Nef bracket multidegrees force $\mathcal{L} \equiv H$; the next inverse direction on $\mathrm{Bl}_5 \mathbb{P}^3$ gives a constant row $(M^{-1})^{\oplus 4} \to M^{-1}$, $M = \mathcal{O}(-h + E)|_C$, $H = \mathcal{O}(3h - E)$. Its Euler relation puts C in a plane transform with $c = 0$, $\lambda = 3$, $d = e + v \geq 4$. Section 5.1.2 excludes ranks 0,1; at F_4, degrees $2, 6, 8, 12$ exclude immersed A_1^2, B_2 planes. The remaining A_2 plane at 1122 has coordinates $qa^3, qb^3, q^2(b^2 - a^2)^3, q^2(b^2 - ca^2)^3$; determinant $9c(c - 1) \neq 0$ modulo the first three products gives tangent dimension 4.

For three labels, $\mathcal{N} = J \oplus M^{\oplus 2}$, $\deg J = 3a + 2x > 0$, $\deg M = -a$, $-a \leq h_6 < 0$. The two highest projections kill J. If $\deg J \leq h_4$, their coefficient inequalities force $a = 1$, $h_6 = -1$, $h_2 = -h_4$, then $h_4 = 1$, $s_3 = 4$, $t = 1$, contrary to $h_6 = 2$ in (5.2). Thus their rank 2 quotient is $M^{\oplus 2}$, and

$$h_2 = 3a + 2x, \quad h_4 + h_6 = -2a, \quad h_4 \leq -a,$$
$$8t = 11a + 10x + p + s_1 + 2s_2 + 3s_3, \quad 4R_F = a + 14x + 3p + 3s_1 + 2s_2 + s_3,$$
$$4R_B = 21a - 10x - 5p - 9s_1 - 6s_2 - 3s_3 - 8s_4, \quad 4h_4 = a - 2x - p - s_1 - 2s_2 + s_3.$$

Without cusps these give $5a - 2x \leq p \leq 21a/5 - 2x$, impossible. For $R_F > 0$, the lowest invariant degrees give generic rank at most 1; Section 5.1.2, with $c = -2$, $\lambda \leq 7/2$, forces $a \leq 2$, $x = -1$. Putting $h_4 = -a - d$, $h_6 = -a + d$, $0 \leq d < a$, gives $t = 2a - 1 + d/2 + s_3/2$, $R_B = 3a/2 + 1 - 3d - p/2 - 3s_1/2 - 3s_3/2 - 2s_4$. For $(a, d) = (1, 0), (2, 0), (2, 1)$ the bounds $t \leq 3/2, 4, 7/2$ contradict $t \geq 11/6, 6, 6$. With $R_F = 0$, $x = -1$, $a \leq 2$, the plane is A_1^2. For $a = 2$, $d = 1$ is inconsistent and $d = 0$ gives $s_2 = 6$, $t = 3$, violating (5.1); for $a = 1$ that equality leaves only (5.4). □

5.2 Fields with two units and the group G_{24}

For $K \neq \mathbb{Q}(i), \mathbb{Q}(\sqrt{-3})$, the full horizontal groups are A_3, B_3, with G_{24} possible only over $\mathbb{Q}(\sqrt{-7})$ [7, 10]. The character fields of $G_{23}, \ldots, G_{27}$ are $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-3})$, and $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$. After G_{24} is excluded, the interior groups are A_4, B_4, D_4, F_4 and transfer restricts rank 2 groups to A_1^2, A_2, B_2 [15, Lemma 2.15]. Their quadratic minimum, descent of $2\mathcal{L}$, Lemma 3.2, and [15, Proposition 2.6] give every hypothesis of Lemma 5.2.

For G_{24}, the horizontal variety is the Jacobian of the Klein quartic with its principal polarization, with generator degrees $1, 2, 4, 7$ [21]. Its unique common fixed point follows from $\det(-g - I) = -1$ for an order 7 element [12, Proposition 2.1(v) in arXiv:2107.03745v2]. The degree 1 section is nonzero there because the quartic is nonhyperelliptic and the tangent representation has no invariant linear form. For this polarization, $\dim \Gamma_{00} = 1$ and the description of Γ_{000} gives a nonzero degree 2 theta projection in the polynomial invariant of degree 4 [17, equations (1.3) and (1.5), Theorems 1.1(2) and 1.3]. A determinant character needs all 21 mirrors, so cannot occur in order 4 or 5; subtract the square of the degree 1 section. For a G_{24} curve, let R_2 count axial inertia 2 and $s = s_4 + s_7$ count cusps according as the degree 14 direction appears in the derivatives of invariant sections of degree 4 or 7. Put $x = g - 1$, $j = 2t - s$. The comparison is $2t = 2x + s + R_2/2$, $a = -24t + 13s - 2x$, $j = 2x + R_2/2$. Hence $a \leq t$ gives $j > 0$, $s \leq 25j + 4x \leq 27j$, $a \leq 14j$. The target associated with the invariant of

greatest degree $Q = E_{14}(4S_4 + 7S_7)$ has degree $-7j - 3s_4 < 0$. Its lower coefficient degrees are $-5j - 3s_4$ and $-s - 4j - 3s_4$, so [15, Lemma 3.5] gives $\dim \mathrm{Hom}(f^*T_X, Q) = 1$. Then [15, Proposition 2.2] requires two coincident labels and $a \geq 21j$, a contradiction. Without cusps, $a + 2g - 2 = -24t$ gives $25a \leq 2 - 2g$, also impossible.

5.3 The primitive Eisenstein groups

Put $K = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$, $\mathcal{O} = \mathbb{Z}[\omega]$, $\delta = 1 + 2\omega$. Initially $6\mathcal{L}$ descends and $a \leq 3t$ [15, Proposition 2.11], by Lemma 3.2. The full horizontal possibilities are $A_3, B_3, G(3,1,3), G(3,3,3), G(6,p,3)$ for $p = 1, 2, 3, 6$, and G_{25}, G_{26}.

For G_{25}, use the E_6 deformation $x^3 + y^3 + z^3 + a_1yz + a_2y + b_2z + a_3$ at $(a_1, a_2, b_2, a_3) = (1, -1/4, -1/4, 5/108)$. With $u = (y - z)/2$, its transverse Milnor basis $1, u, u^2, x, xu, xu^2$ gives an isomorphic Kodaira–Spencer map [24]. Put $s = (a_2 + b_2)/2 + 1/4$, $w = (a_2 - b_2)/2$, $c = a_3 - 5/108$. The regular order 3 element has fixed weights $6, 9, 12$, centralizer G_{25}, and linear terms $(P_6, P_9, P_{12}) = (-3s, 2w, c + s/3)$ [25]. Reversing weight 9 gives G_{26}, of degrees $6, 12, 18$. Its two models have first theta minors $\left(\begin{smallmatrix} -2/3 & 0 \\ 1/9 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1/9 & 1 \\ 1/6 & 0 \end{smallmatrix}\right)$. For G_{25} the normal differential is $\mathrm{diag}(q^2, q^2, q^3) \left(\begin{smallmatrix} b_6 & 0 & * \\ 0 & b_9 & 0 \\ 0 & 0 & b_{12} \end{smallmatrix}\right)$, $b_6 b_9 b_{12} \neq 0$. Weight preserving equivariant changes preserve these minors. The degree 1 section is nonzero at the three G_{25} points $(-\zeta^{-1}/4, -\zeta/4, 5/108)$, $\zeta^3 = 1$, and at the unique G_{26} origin obtained by the involution and degree 3 isogeny. Thus the theta projections have the stated orders, trivial fiber characters, and width 1.

In the following calculations $\sigma = u/2 + 2v/3 + 5w/6$, with $u, v, w \geq 0$ integers, counts product contributions. For G_{26}, s_1, s_2 count 1234, 1346 cusps and $j = 6t$. There is no irreducible larger group, and

$$j/3 = 2g - 2 + s_1 + s_2 + \sigma, \quad a = 10s_1 + 14s_2 + \sigma - 19j/3, \quad h = -3j + 4s_1 + 6s_2.$$

Since $j \in \mathbb{Z}$, the first equation gives $3\sigma \in \mathbb{Z}$. These equations force $g = 0$, $h < 0$, $a = s_1 + s_2 + 2h + 2$. The lower residual degrees are at most $(h - j)/2, h/3$, so the highest Hom space has dimension 1 and $a \geq -3h$ [15, Lemma 3.5]. Equivalently, $21s_1 + 13s_2 + 39\sigma \geq 82$ and $24s_1 + 14s_2 + 45\sigma \leq 92$. The exhaustive bounds are

(s_1, σ)	$(0,0)$	$(1,0)$	$(2,0)$	$(0,2/3)$	$(1,2/3)$	$(0,1)$
lower s_2	7	5	4	5	3	4
upper s_2	6	4	3	4	2	3.

For G_{25}, let R count its only irreducible larger group G_{32}. The comparison, retaining the actual finite character, is

$$2t = 2g - 2 + s + 5R/6 + \sigma, \quad a = 8s - 29t + 28R/3 + \sigma,$$
$$\deg(E_6, E_9, E_{12}) = (-6t, -9t - R/2, -12t), \quad h = \deg E_{12}(3S + 3D_R) = 6 - 6t + R/2 - 3\sigma.$$

It gives $g = 0$, $h \leq 0$. For the matrix at G_{32}, extract B by Section 4.2, divide each column by its nonzero first entry, and set $A_{ij} = B_{i+1,j+1} - B_{i+1,1}B_{1,j+1}$. The matrix $M = \mathrm{diag}(v^2, v^3, v^4) A \,\mathrm{diag}(1, 1, v^{-1})$ and its Schur complement D have entries $A_{11} = -650/297$ and $D_{11} = (-19661200/649539)/A_{11}$. Their determinants are $\det A = -19489164500000/1162261467$ and $\det D = \det A/A_{11}$. The degree 6 coefficient has residual degree $-4t + 2 + R/6 - \sigma < 0$. The other coefficients have zeros $(2,3)$ at cusps and $(3,2)$ at G_{32}, with $D_{11}(b_{12}/v^2)(0) = D_{12}(b_9/v^3)(0)$. Surjective evaluation at the distinct R points gives $\dim \mathrm{Hom}(\mathcal{N}, E_{12}(3S + 3D_R)) = \max(\eta + 1, 0) + 1$, $\eta = -t + 2 - R/3 - \sigma$. Substitution in the preceding equations is exhaustive under the following conditions.

	integer inputs and bounds	$(a, h, \dim \mathrm{Hom})$
$\eta \geq 0$	$0 \leq R \leq 10$, $0 \leq s \leq 4$, $0 < 6t < 20$, $6t+R$ even, $1 \leq a \leq 3t$	$(4,-1,2)$, $(2,-1,2)$, $(3,0,3)$
$\eta < 0$	The same bounds, $a \geq -3h$, $R+39\sigma/16 \leq 3$	No solution.

The outputs $(4,-1,2), (2,-1,2)$ contradict [15, Proposition 2.2]. For $(3,0,3)$, the identities $M_i = L_i^2 \otimes \det(V)^{-1}$ and $a \equiv \deg V \pmod 2$ make the contribution $(m-1)(e+1)$ of each negative coincident component even when a is odd. This excludes the remaining Hom dimension, every G_{25} cusp, and the G_{32} points through their actual parabolics.

5.4 The sixth roots of unity

Rains's 16 monomial actions come from $E : z^2 = y^3 + x^6$, with $d_2 \mid 2$, $d_3 \mid 3$, $d_6 \mid 6$; no translation isogeny occurs because $1-\zeta_6$ is a unit [21, Theorem 3.6, Example 3.11]. Put $d_0 = \gcd(2d_3, 3d_2)$, $g_0 = \gcd(d_6, d_0)$, $q = 6/\operatorname{lcm}(d_2, d_3, d_6)$, $p = 6/d_6$. The full group is $G(6,q,3)$, its subgroup with trivial fiber character is $T = G(6,p,3)$, the primitive degrees are $(d_6, 2d_3, 3d_2, 6)/g_0$, and the width is $\kappa = d_0/g_0$. For $p > 1$ put $B = (d_6 + 2d_3 + 3d_2 + 6)/d_0$, $\nu = 6/d_0$, $\eta = d_6/d_0$, and $\ell = 3$ for $d_0 = 1$, $\ell = \nu$ otherwise. Let R, σ count $G(6,p,4)$ and product contributions, and let s_0 count $d_0 = 1$ ends, where the curve has a semicubical cusp and saturated tangent bundle $T_C(S_0)$. Projection of the inverse differential to the product invariant and to the elementary symmetric invariant of greatest degree gives simultaneous everywhere surjective quotients with $h_P = -18t/p + R/p + \sum \eta_c$, $h = -12t + R + \sum \nu_c$, and

$$\begin{gathered} 2t = 2g - 2 + s + 5R/6 + \sigma, \quad a = \sum B_c + (17/6 + 1/p)R + \sigma - (20 + 18/p)t, \\ ph_P = h - 6t, \qquad a = (2p+1)h_P + 18t + 2 - 2g - 2s_0. \end{gathered} \tag{5.5}$$

The other coefficient degrees for the first of these two targets are $h_P + 6t - R - \sum \ell_c$ and $h_P - h$.

To recover the local inputs, use $S_1 = \sum w_i^6$, $S_2 = \sum_{i<j} w_i^6 w_j^6$, $P = (w_1 w_2 w_3)^{6/p}$. The full normalizer preserves their lines because coordinate axes contain more mirrors than other intersection lines. Primitive alternatives are excluded by the 18 order 2 exchange reflections, compared with zero and nine in G_{25}, G_{26}. The two theta projections onto (dS_1, dS_2) have matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; an equivariant change $\xi_i = bw_i + O(w_i^7)$ preserves the required zeros. At larger points, $v = u^6$, $dU_2 = v\,dS_1 + dS_2$, $dU_3 = v\,dS_2$, $dU_0 = u^{6/p} dP$. A frame character ζ^{-r} contributes $r/6$ after the allowed pole, giving the fractional η in (5.5).

The cases $p = 1, 2$ reduce to the following inequalities. For $p = 1$, $(A, b) = (4,1), (7,2), (10,3), (17,6)$, all widths are 1, and the 2366 tangent correction cancels the removed determinant order. Put $j = 6t$, $V = \sum(A_c - 1 - 2b_c) \geq 2\sum b_c/3$. Then $g = 0$, $h = -3j + \sum b_c + R < 0$, $a = V + 2h + R + 2$, $8h \leq 3a - 6j - R - 6$, hence $11a \geq 6j + R + 6$, contradicting $a \leq j/2$. For $p = 2$, $(B, \nu, \ell) = (14,6,3), (17/2,3,3), (6,2,2), (7/2,1,1)$. Here $g = 0$, $a = 4h_P + 9t + R/2 + V_2 + 2$, $V_2 = \sum(B - 1 - 2\nu) > 0$, so $h_P < -3t/2$, $h = 2h_P + 6t \geq 0$. Both residual degrees are negative; the one dimensional Hom condition gives $a \geq -3h_P > 9t/2$.

For $p = 6$, $13h_P + 11t + 2g + 5R/3 + 2\sigma \leq 2$ gives $h_P < 0$. If $g \geq 1$, the negative residual degrees and $a \leq t$ contradict $a \geq -3h_P \geq 3t$, so $g = 0$. If $h < 0$, the simultaneous rank 2 quotient gives $h_P = h = -a$, $t = 5a/6$, $s_0 = 1 + a/2$. The bound $\sum \nu \leq 3s + 3s_0$ gives $5a/2 + 3R/2 + 3\sigma \leq 9$, hence $a = 2$, $s_0 = 2$, impossible for a line or immersed conic. If $h \geq 0$, put $h_P = -b$. The GIT kernel and negative residual degrees give $h \geq b$, $a = 3b + 2v$, $0 \leq v \leq h/4$, $s_0 = b + 1 + 3h/2 - v$. The five outside sections give a base point free map $\gamma : \mathbb{P}^1 \to \mathbb{P}^3$ of degree $b+v$ whose differential vanishes at least $s_0 - v$ times. For span dimension ρ, its Wronskian gives $\rho(s_0 - v) \leq (\rho+1)(b+v-\rho)$, $(3\rho-1)h/4 \leq b - \rho(\rho+2)$. This contradicts $h \geq b$ for $\rho = 2, 3$; a line germ cannot be semicubical.

For $p = 3$, $a \le 26(1-g) - 23R/3 - 12\sigma$ gives $g = 0$. The identity $a = 6h_P + 12t + R/3 + \sum(\nu - 21_{d_0=1}) + 2$ gives $h_P < -3t/2$. The residual degrees are $-2R/3 + \sum(\nu/3 - \ell) < 0$ and $-2h_P - 6t \le 0$. Equality in the second gives $h_P = h = -a$, $a = 3t$, three coincident labels, and a kernel line of degree $3a - 2 - s_0$ mapping to degree $a - 3s_0$, hence $a + s_0 \le 1$. The remaining triples $(d_2, d_3, d_6) = (2,1,2), (1,3,2), (2,3,2)$ have full groups/widths $(G(6,3,3),1), (G(6,1,3),3), (G(6,1,3),3)$. The first has no order 6 reflection; in the others its fiber character is nontrivial on the connected fixed torus. Transfer excludes every actual interior order 6 reflection [15, Lemma 2.15]. Thus $a = 1$, $t = 1/3$, $s_0 = 0$, and the plane has generic group $G(6,3,2)$: Lemma 3.2 excludes ranks 0,1 and the other rank 2 groups by their quadratic minimum.

5.5 The fixed surface and its full lattice

Let $Z \simeq \mathbb{P}^2$ have generic group $T = G(6,3,2)$. The normal representation invariants have degrees 4,6; along the $G(6,3,3)$ curve D_6, their inverse coarse frames are $u^{-2}e_4, v^{-1}e_6$, $v = u^6$. The actual line bundles satisfy $Q_4^{\otimes 3} = \mathcal{L}^{-12}|_Z \otimes \mathcal{O}_Z(D_6)$ and $Q_6 = \mathcal{L}^{-6}|_Z \otimes \mathcal{O}_Z(D_6)$, including finite characters because the punctured plane has unchanged Picard group and trivial fundamental group. Hartogs extension of the nonzero inverse determinant gives $Q_4 = Q_6 = \mathcal{O}_Z(-1)$, $\lambda = 1/3$, $\deg D_6 = 1$, orbifold Euler number $1/3$. Every cusp lies on D_6: the connected T fixed torus contains the origin and has constant fiber character, so lifting the entire group gives $G(6,3,3)$ with axial inertia 6 and actual width 1 or 3.

Lemma 5.3. The full effective arithmetic group of Z is conjugate to the subgroup H_2 specified in the following proof. Its mirror of order 6 has signature $(0;6,6;1\text{ cusp})$ and coarse degree 1.

Proof. Put $\pi = 1 - \omega$, $h_0 = -\left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$. The four possibilities are the projective full unitary groups $M_{\epsilon\eta}$ of $L_{\epsilon\eta} = \mathcal{O}(2^\epsilon \pi^\eta)^{-1} e_1 \oplus \mathcal{O}e_2 \oplus \mathcal{O}e_3$, $\epsilon, \eta \in \{0,1\}$ [26, 19]. For principal Euler number $A/24$, normalizer index n, and target index j, the volume formula gives $jA = 8n$. The bound $n \le 3^{1+|I|}$, split Iwahori factors ≥ 456, and inert factors $p^2 - p + 1 = 3, 21, \ge 111$ at $2, 5, \ge 11$ force $I = \varnothing$, $n \le 3$, $A \le 24$, and allow only the vertices at 3 and choices at 2. The determinant of $\mathrm{diag}(1,\omega,1)$ is not a cube, giving $n = 3$. The Euler numbers are $1/72, 1/72, 1/24, 1/24$, hence target indices $24, 24, 8, 8$. Each group has one cusp since the class number and its 3 torsion are 1 and $I = \varnothing$ [27, Theorem 4.8].

Write E_{ij} for matrix units of size 3. For M_{00} use $P = I + E_{12} - \omega E_{13} + (\omega - 1)E_{22} + \omega E_{23}$, $Q = I + E_{12} - \omega E_{13} - 2E_{22} - E_{23}$, $R = -E_{13} - E_{22} - E_{31}$, $J = RP$, $\sigma = QP^{-1}$. Its complete presentation is $J^3 = \sigma^6 = (J\sigma^{-1}J)^4 = 1$, $[\sigma, (J\sigma^{-1}J)^2] = 1$, and cusp group $\langle P, Q\rangle$ [4, Theorem 5.9, Proposition 5.11]. For M_{01} use $S = I + E_{12} + \overline{\omega}\delta E_{13}/3 - 2E_{22} - E_{23}$, $T_0 = I - \delta E_{13}/3$, $I_s = -\delta E_{13}/3 - \omega E_{22} - \delta E_{31}$. Putting $j = S^{-1}I_s$, $r = T_0^{-1}I_s$, $z = jr^{-1}j = I_s$, its complete presentation is $j^3 = r^3 = z^{12} = 1$, $[r, z^2] = 1$, and cusp group $\langle z^2, S\rangle$ [31]. These presentations come from fundamental domains.

Set $H_\eta = M_{0\eta} \cap M_{1\eta}$. Reduction modulo 2 has image of order 216, transitive on nine isotropic lines; H_η fixes the first line and has indices 9,3. Reidemeister–Schreier rewriting of this nine point action gives ten generators and 36 complete relations. For $U = I + \delta E_{13}/2$, generators of M_{10} are $h_1 = J^{-1}\sigma J$, $h_2 = \sigma$, $h_3 = (J\sigma J^{-1})\sigma^3(J\sigma^{-1}J^{-1})$, U; those of M_{11} are $k_1 = S^{-1}$, $k_2 = r^{-1}I_s r$, U. Their right coset representatives over H_η are $1, U, UC$, with $C = h_3, k_2$ respectively. To obtain a complete presentation, rewrite $r_i g = w_{i,g} r_{\kappa(i,g)}$ by selecting the unique representative whose quotient fixes the first isotropic line modulo 2. Enumerate $w_{i,g}$ in h_1, h_2, h_3 or k_1, k_2 until the projective matrices agree. The nine point presentation verifies that these generate H_η, guaranteeing termination; the rewritings and its 36 relations present $M_{1\eta}$.

Enumerate transitive actions by extending the first undefined generator image, propagating inverses and all relators, rejecting contradictions or excess points, and relabeling breadth first from every base point. A branch assigns at most n images for each of the finitely many generators, where $n = 24$ or 8; each unassigned image has at most n choices. Propagation only fills these entries or rejects the branch, so the enumeration terminates without a bound on word length or running time. Every complete transitive action extends a branch, and relabeling from every base point identifies precisely simultaneous conjugacy, so the procedure is exhaustive. At index 24 in $M_{0\eta}$, require a conjugate of σ and exclude $J, (\sigma J)^3$ or j respectively; their isolated full stabilizers of orders 3,4 contain no reflection [3, Section 7, Tables 9–10]. This leaves eight and six actions. At index 8 in $M_{1\eta}$, $\Delta = \Lambda \cap H_\eta$ satisfies $3[H_\eta : \Delta] = 8[\Lambda : \Delta]$ and $[\Lambda : \Delta] \le 3$, hence the indices are 8,3. There are 20,12 eight point H_0, H_1 actions. Induce them to $(a, i)g = (aw_{i,g}, \kappa(i, g))$ on 24 points; extensions are the 64 blocks of size 3 through the base point, one point per fiber. Testing every generator and inverse translate for equality or disjointness and canonically relabeling leaves four actions for each η.

For $\epsilon = 1$ the full cusp groups are $\langle P, Q, U\rangle$, $\langle z^2, S, U\rangle$. Their horizontal module is $\mathcal{O}$ and central period is $\delta/(2^\epsilon 3^\eta)$. Since $1 + \overline{\omega}$ is a unit, every primitive order 6 cusp element is conjugate to $\sigma^{\pm 1}$. Requiring a σ fixed point in every cusp orbit leaves the full $4, 4, 2, 2$ cases.

For the full stabilizer tests put $D = \operatorname{diag}(1, 1, -1)$, $A = D^{-1}\sigma D$, $J_+ = D^{-1}JD$, $R_+ = D^{-1}RD$, $r_2 = J_+AJ_+^{-1}$, $r_3 = J_+^{-1}AJ_+$, $b = r_2r_3r_2^{-1}$, $d = r_2J_+^2$. The standard point groups at $[-1, 0, 1]$, $[\omega^2, 0, 1]$ are $\langle A, R_+\rangle$, $\langle A, b, d\rangle$, of orders 12,72. Enumerate their projective matrices fixing each coset and test local reflections by $\operatorname{rank}(M - \alpha I) = 1$, $Mz = \alpha z$. For mirror orbit length d and generic inertia $k \mid 6$, the effective index is $dk/6$. The standard full norm 1 mirror normalizer $\langle A, R_+, b\rangle$ has kernel μ_6 and effective group $\Delta(2, 6, \infty)$ [18, Section 3]. The M_{01}, M_{11} normalizers have full effective group $\mathrm{PSL}(2, \mathbb{Z})$: they contain its generators, elliptic orders are $2, 3, 6$, order 4 fails the valuation at 2 of $(\operatorname{tr} M)^2 = 2 \det M$, and the minimal nonuniform covolume is $1/6$. Thus $-\chi(C) = 2\mathcal{L} \cdot C = 2 \deg C/3$ gives degrees $dk/12, dk/24$ respectively. The complete results are as follows; paired orders give the full point group and its reflection subgroup.

group	cases	exclusion or surviving case
M_{00}	4	Point order 36; orders $(12, 4), (6, 1)$; H_2.
M_{01}	4	Mirror degrees $1/2, 3/2, 3/2, 3/2$.
M_{10}	2	Mirror degree 2; point order 36 on $\operatorname{diag}(1, 2)$.
M_{11}	2	Mirror degrees $2, 1/2$.

The deficient point groups in M_{00} violate the required order 72 or the Chevalley–Shephard–Todd theorem. For the binary lattice, $|a|^2 + 2|b|^2 = 1$ gives full group μ_6^2, excluding the required effective $G(6, 1, 2)$ of order 72. All fractional or larger degrees contradict $\deg D_6 = 1$. The H_2 mirror has effective index 2 in $\Delta(2, 6, \infty)$, proving its signature and degree. □

5.5.1 The linear group and its preserved lattice

Use negative lines, $h_3 = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$, and

$$J = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -\omega & \omega \\ 1 & 1 & \omega \end{pmatrix}, \quad A = \operatorname{diag}(1, -\omega^2, 1), \quad C = \omega I, \quad S = JA^{-1}J.$$

The complete central presentation is $J^3 = C$, $A^6 = 1$, $S^4 = C^2$, $[A, S^2] = 1$, $C^3 = [J, C] = [A, C] = 1$. The subgroup H_2 is the inverse image of the stabilizer of 1 in the right action

defined by the following disjoint cycles, with C acting trivially:

$$\begin{aligned} J &= (1,2,3)(4,6,7)(5,9,10)(8,16,12) \\ &\quad (11,20,21)(13,22,14)(15,23,17)(18,24,19), \\ A &= (2,3,5,11,8,4)(6,12,13)(7,14,18,9,17,15) \\ &\quad (10,19,20)(16,22)(21,24). \end{aligned}$$

Choose the first breadth first word t_i from 1 to i, in alphabet order J, A, J^{-1}, A^{-1}. The 72 edges are $e_{i,g} = t_i g t_{ig}^{-1}$ for $g = J, A, C$; inverse edges are $e_{i,g^{-1}} = e_{ig^{-1},g}^{-1}$. The 23 tree edges and seven central relations rewritten at all 24 points give 191 relations, of full rank with nonunit Smith entries $6, 6$, hence abelianization $(\mathbb{Z}/6\mathbb{Z})^{\oplus 2}$. Write $\chi(e_{i,g}) = \omega^{x_{i,g}}$ and impose $x_{i,C} = 1$. The conditions are value 1 on norm 1 reflections of orders 6 and 2, value ω^2 on those of order 3, value 1 on norm 2 reflections of order 2, and agreement with the degree 4 normal character at the two full points of order 72. To reproduce them, put $r = C^{-1}S^2$, $b = (JAJ^{-1})(J^{-1}AJ)(JA^{-1}J^{-1})$. Take all orbits of $\langle A, r, b\rangle$ and $\langle A, S\rangle$ in the displayed action and rewrite the conjugates of the appropriate inertia powers from an orbit representative. At base points 1 and 23 impose exponents $0, 0, 1$ on A, $JAJ^{-2}AJ^2A^{-1}J^{-1}$, and JAJ, conjugated by t_i. The 191 relations have rank 70 over $\mathbb{F}_3$; imposing $\chi(C) = \omega$ gives rank 71, and $\chi(A) = 1$ determines a unique character. Its values $\chi(e_{i,g}) = \omega^{x_{i,g}}$ are $x_{i,C} = 1$ for every i, and

$$\begin{aligned} x_{i,J} &= 1 \quad \text{for } i \in \{2,6,9,12,17,20,22,24\}, \\ x_{i,A} &= 1 \quad \text{for } i \in \{2,16,17,20,22\}, \\ x_{i,A} &= 2 \quad \text{for } i \in \{11,14,19,24\}, \end{aligned}$$

with all remaining exponents zero. Substitution in the rewritten relators and the preceding inertia words verifies every stated local value. Evaluate any word by summing its Schreier exponents.

On the fixed space W and its positive normal space, the actual full normalizer is

$$\begin{aligned} \rho(M) &= \operatorname{diag}(\chi(M)^{-1}M, (\det M)^{-1}, 1), \\ \iota(t) &= (\det t)\operatorname{diag}(I_3, t), \quad t \in G(6,3,2). \end{aligned}$$

Every block differs from $\rho(M)$ by $\iota(T)$, since the determinant condition first gives a scalar in μ_6 and the character conditions reduce it to $\chi(M)^{-1}$. Both order 6 elliptic points of the mirror therefore have actual ambient group $G(6,3,4)$. The parabolic classification excludes primitive larger groups. A product would require a reflection of order 6, already excluded. Standardize one such full point group; no relative standardization of the second point is assumed. Put $M_1 = A$, $M_2 = JAJ$, $M_3 = JA^{-1}JAJA^2J^{-1}$. Their traces are $\omega + 3, -\omega, -\omega$. Solving $XM_i = M_iX$ gives commutant KI, and the corresponding intertwiner equations have dimension 1 over K. The basis $(\omega^2, 0, 1), (0, 1, 0), (-\omega, 0, 1)$ has Gram matrix $\operatorname{diag}(-1, 1, 1)$. For the first full $G(6,3,4)$ group, let P_i project to a positive axis and put $F_i = -I + 2P_i$. With $\zeta = 1 + \omega$, let D_{ij} act by ζ, ζ^{-1} on the i, j axes and by 1 on the other axes. The identity

$$P_i = \omega^2(D_{ij} - I)(D_{ij} - \zeta^{-1}I) - \omega(F_i + I)$$

places P_i in the integral group algebra. For $B_3 = \left(\begin{smallmatrix} \omega^2 & 0 & -\omega \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$, the ideals generated by entries $12, 13$ and by entries $21, 31$ of the 72 matrices $B_3^{-1}e_{i,g}B_3$ are both $\delta\mathcal{O}$. Divisibility of all generators and the entries $\omega - 1, -2(\omega - 1), -\delta, 2\delta$ of $B_3^{-1}M_3B_3$ in positions $12, 13, 21, 31$ prove both ideal equalities for the original full group. The trace and commutator calculation above gives a conjugacy over K, and the actual monomial interchanges connect the normal axes to give the form $h_5 = \operatorname{diag}(-1, 1, 1, 1, 1)$. The same full Γ preserves an $\mathcal{O}$ lattice by [15,

Proposition 2.11]. Integral projectors from the first $G(6,3,4)$ group split this lattice as $d\mathcal{O}\oplus\mathcal{O}^4$. The ideals generated by the entries in its two off diagonal blocks are $\delta\mathcal{O}$, so $\delta\mathcal{O}\subset d\mathcal{O}\subset\delta^{-1}\mathcal{O}$. The three possible lattices are exchanged by duality; the unique intermediate module in $\mathcal{O}/\delta^2\mathcal{O}$ is the middle lattice. Thus every isometry of any of them preserves the middle lattice, and the original full group satisfies

$$\Gamma\subset \mathrm{SU}(h_5,\mathcal{O}).$$

5.5.2 An actual reflection of order 6

For a positive vector p in the already fixed form h_5, put

$$P_p=\frac{pp^*h_5}{p^*h_5p},\qquad B_p=-I+2P_p.$$

The following basis is integral and has determinant 1:

$$v=(\delta,1,1,1,0),\quad q_1=(1,0,\omega^2,-\omega,0),$$
$$q_2=(1,0,-\omega,\omega^2,0),\quad q_3=e_4,\quad t_0=e_1+\omega v.$$

Write $Q_v=(v,q_1,q_2,q_3,t_0)$, $B=\mathrm{diag}(B_3,I_2)$, $s=(1-\omega,2\omega^2,-1-\omega,0,0)$, and

$$G=B^{-1}\,\mathrm{diag}(\omega^{-1}M_3,(\det M_3)^{-1},1)B,$$
$$T_n=\mathrm{diag}(I_3,-I_2),\quad F_0=\mathrm{diag}(1,V_0),$$
$$V_0=\begin{pmatrix}0&\omega&0&0\\-\omega&0&0&0\\0&0&\omega&0\\0&0&0&1\end{pmatrix}.$$

Then $B_s=F_0T_nGF_0^{-1}$ belongs to the same Γ. Let $F=\mathrm{diag}(1,\left(\begin{smallmatrix}0&1&0\\0&0&1\\1&0&0\end{smallmatrix}\right),1)$, and let Z be minus the interchange of the first two positive axes. For $X=FB_sZ^{-1}$ and $Y=FXF^{-1}$, the matrices in the Q_v basis are $N((-1,1,0),-1)$ and $N((-\omega,-1-\omega,0),-1)$, where

$$N(b,u)=\begin{pmatrix}1&-\bar b^t&u\\0&I_3&b\\0&0&1\end{pmatrix}.$$

They fix q_3 and change q_1+av to $q_1+(a+1)v$ and $q_1+(a+\omega^2)v$. Thus they remove any $a\in\mathcal{O}$. Every full cusp element acts monomially on the unimodular positive horizontal lattice; its scalar is in μ_6 and a trivial horizontal action would give $\lambda^5=1$, hence $\lambda=1$. If that full action preserved q_3, it would split the horizontal representation as $2+1$, giving positive second Betti number by the circle bundle Gysin sequence. Thus some actual cusp element moves q_3 to $u(q_i+av)$, $i=1,2$. Conjugation of B_{q_3}, the equality $Zq_2=(1+\omega)q_1+\omega v$, and X,Y give $B_{q_1}\in\Gamma$. For $r_0=(1,1,1,0,0)$, the actual element

$$F_1=-\,\mathrm{diag}\left(1,\begin{pmatrix}0&\omega&0&0\\0&0&-\omega^2&0\\1&0&0&0\\0&0&0&1\end{pmatrix}\right)$$

satisfies $F_1q_1=-r_0$. With $D=\mathrm{diag}(1,1,\omega,\omega^2,1)$ and $u_0=(2\omega,2\omega,\omega^2,0,0)$, the identity $P_{DB_{r_0}e_2}=P_{u_0}$ gives $B_{u_0}\in\Gamma$. Finally put $w_3=JA^{-2}J$, $M_3'=w_3A^2w_3^{-1}$; the character above gives $\chi(M_3')=\omega^2$. The actual element

$$T_3=B^{-1}\,\mathrm{diag}(I_3,\omega^2,\omega)\,\mathrm{diag}(\omega M_3',(\det M_3')^{-1},1)B=\omega I+(\omega^2-\omega)P_{u_0}$$

has product with B_{u_0} whose polar eigenvalue divided by its mirror eigenvalue is $-\omega$. All displayed products are exact identities over K.

This actual reflection contradicts its prior exclusion and eliminates the three remaining sixth root models.

5.6 The cubic groups and the last cusps

Only $G(3,1,3), G(3,3,3), A_3, B_3$ remain. Their normal representation invariants give $\lambda \geq 1/2$ by Lemma 3.2, hence $a \leq 2t$ by [15, Proposition 2.6]. For $G(3,1,3)$, the cusp degrees are $1111, 1133, 1333$, all widths are 1, their sums are $4, 8, 10$, and highest poles are $1, 3, 3$. Let A_c, b_c be the sum of generator degrees and highest pole order at cusp c. Put $j = 3t$, let R count $G(3,1,4)$, $B = \sum b_c$, and $V = \sum(A_c - 1 - 2b_c)$. The comparison gives $g = 0$, $h = -3j + R + B$, $a = V + 2h + R + 2$, $V \geq B/3$, $(7/3)B \leq 7j - 3R - 2$. The lower residual degrees are negative, the first being $-2j + B - V$; thus the target associated with the invariant of greatest degree has $\dim \mathrm{Hom}(\mathcal{N}, Q) = 1$ and $a \geq -3h$. The finite conditions and their complete outputs are as follows; all counts and j are integers, and outputs are ordered as $(j, R, s_1, s_2, s_3, \sigma; a, h)$.

inputs and equations	outputs
$1 \leq j \leq 28$, $0 \leq R \leq 6$, $0 \leq s \leq 20$, $\sigma = u/2 + 2v/3 + 5w/6$; $2j/3 = -2 + s + 2R/3 + \sigma$, $a = \sum A_c + 11R/3 + \sigma - 20j/3$, $h = -3j + R + \sum b_c$; $1 \leq a \leq j$, $-a \leq h < 0$, $a \geq -3h$	$(6,0,1,4,1,0;6,-2)$, $(7,1,0,5,1,0;7,-2)$.

Both have $a = 3t$, contradicting $a \leq 2t$. Transfer now excludes actual order 3 reflections; $2\mathcal{L}$ descends, and the remaining quadratic minimum gives $\lambda \geq 1$, $a \leq t$.

For $T = G(3,3,3)$, let R count $G(3,3,4)$. Its degrees are $3, 3, 6$; the full normalizer may mix the first two directions, and every cusp is 1113 of width 1. Write $P_3 = w_1 w_2 w_3$, $S_1 = \sum w_i^3$, $S_2 = \sum_{i<j} w_i^3 w_j^3$. At larger points $v = u^3$, $dU_0 = u\, dP_3$, $dU_2 = v\, dS_1 + dS_2$, $dU_3 = v\, dS_2$. At cusps the theta projections are $q\, dP_3$, $-q\, dS_1/3$, $q^3(dS_1 + dS_2)$. The projection of the inverse differential to the coordinate of greatest invariant degree is therefore $3q^{-1}dU_2 + q^{-3}dU_3$, including its lower term, and gives an everywhere surjective quotient $\mathcal{N} \to Q_6$ after pole 3. For $K_3 = \ker(\mathcal{N} \to Q_6)$, the low degree directions have pole at most 1 on a finite cover trivializing their representation. Consequently every line subbundle $J \subset K_3$ satisfies $\deg J \leq b = -3t + R + s$. The quotient degrees are $\deg K_3 = 2b - 2R/3$ and $h = -6t + R + 3s$. The tangent comparison gives $2t = 2g - 2 + s + 2R/3 + \sigma$. The normal comparison gives $a = 6s + 3R + \sigma - 14t$. These force $g = 0$. With $j = 6t$, the exhaustive conditions and remaining cases are

inputs and equations	nine outputs
$1 \leq j \leq 33$, $0 \leq R \leq 7$, $0 \leq s \leq 13$, $j/3 \in \mathbb{Z}$; $\sigma = j/3 + 2 - s - 2R/3$, $a = 6s + 3R + \sigma - 7j/3$, $h = -j + R + 3s$, $b = -j/2 + R + s$; $a \in \mathbb{Z}$, $\sigma \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, $1 \leq a \leq j/6$; for $R = 0$ also $b \in \mathbb{Z}$	Three with $R = 0$; two with $R = 3$, integral t; $(a,t,b) = (1, 3/2, 1/2)$; three with $(a,t) = (2,2), (1,5/2), (2,7/2)$.

For $R = 0$, the unramified representation is projectively trivial on $\mathbb{P}^1$, so $K_3 = \mathcal{O}(b)^{\oplus 2}$; the three negative rows contradict [15, Proposition 2.4]. The row $(1, 3/2, 1/2)$ has a positive normal line of degree 1 in K_3. There is no row with $s = 0$: for $R = 0, 3, 6$, $[(7R/3 + 2)/13, (7R/3 + 2)/12)$ contains no half integer.

The full projective normalizer has order 216, generated by T, $\mathrm{diag}(\omega, 1, 1)$, $\delta^{-1}\left(\begin{smallmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{smallmatrix}\right)$. Images of a four point frame in the nine mirror lines prove completeness. Its low degree representation, generated by $D_2 = \mathrm{diag}(\omega, 1)$, $F_2 = \delta^{-3}\left(\begin{smallmatrix} -3 & 1 \\ 18 & 3 \end{smallmatrix}\right)$, has order 24 and $F_2^2 = -I$. Direct multiplication shows that a triple of the four conjugates of D_2^2 has product I only when its factors agree. For integral t, $\mathcal{O}(t)$ cancels the actual central μ_2 character of $2\mathcal{L}$; no such

cancellation is asserted for half integral t. Thus the $R = 3$ integral rows have a global lowest line of degree $-3t + R/3 + s$. One is less than $-a$; the other places the positive line of degree $3a - 2 = 7$ in K_3 with $b = 0$.

For the last three rows, every immersion of H degree 1 or 2 is a line or conic in a coincidence plane, or moves one label with six fixed. The splitting $V = \mathcal{O} \oplus \mathcal{O}(k)$ gives $a = (7-2m)k+2\sum b_i$, $k, b_i \geq 0$. A conic's positive normal line of degree 4 maps trivially to Q_6 and exceeds the K_3 bound. At a 1113 cusp the exact torus and nine directions are

$$A = E^3/K_0, \quad E: \ x_1^3 - x_2^3 - x_3^3 = 0, \quad K_0 = \{(k_i) \in E[1-\omega]^3 : \sum k_i = 0\},$$

$$(0, \omega^j, -\omega^{-j}), \quad (1, 0, \omega^j), \quad (1, \omega^j, 0), \qquad j = 0, 1, 2.$$

They are the three degree 1 theta section values at the fixed points represented by $(u + k_i)$, $u \in E[3]/E[1-\omega]$, $\sum k_i \in E[1-\omega]$. At least two values are nonzero, proving trivial fiber character and width 1. The directions lie on a smooth cubic, so at most three lie on a line. With zero, one, or two triple points, there are at most $7, 3, 0$ complete moving label curves and $0, 1, 2$ coincidence planes. These account for at most $7, 6, 6$ directions, a contradiction. Thus neither $G(3,3,3)$ cusps nor $G(3,3,4)$ points occur.

Only Coxeter horizontal groups remain. Their normal characters may have an unramified μ_3 factor of degree 0, but at a cusp $1 = b^5 \det F$, $b \in \mu_6$, $\det F = \pm 1$ gives $b = \pm 1$. There is no extra cusp inertia, so Lemma 5.2 applies and proves Proposition 5.1.

6 Seven points over the Gaussian field

In this section, $X = X_7$, $H = -K_X$, and $K = \mathbb{Q}(i)$. We assume that $X = (\mathbb{B}^4/\Gamma)^{\mathrm{BB}}$ for a nonuniform arithmetic lattice. All fixed quotients are formed with the full normalizer in this same group Γ. We use G_j for the Shephard–Todd group numbered j, and A_j, B_j for the usual Weyl groups.

6.1 Reflections of order four

Lemma 6.1. No smooth cusp has full horizontal group $G(4,1,3)$.

Proof. For the full fixed curve [15, Lemma 2.17], put $a = \deg f^*H, t = \deg f^*\mathcal{L}, j = 4t \in \mathbb{Z}_{>0}$; Lemma 3.2 gives $a \leq j/2$. The upper groups $G(4,1,4), T \times \mu_2, T \times \mu_4$ have counts R, u, v, $\sigma = u/2 + 3v/4$ [10, Theorems 2.1, 3.6]; the maximal end group T has normalizer $\mathbb{C}^*T$, preserving its three invariant directions of degrees $4, 8, 12$ and their characters. The actual degrees and inverse corrections are [21, Theorem 3.6 and Example 3.10]

degrees	1111	1122	1222	1244	1444
(ν_1, ν_2, ν_3)	$(1,1,1)$	$(1,2,2)$	$(2,2,2)$	$(2,4,4)$	$(4,4,4)$.

The full least period has width 1, by the degree 1 axis section's trivial fiber character; raw 2244 loses exactly its kernel μ_2. For $X_0 = \prod x_{1j}, Y = \prod y_j$ on $E: x_1^4 - x_2^4 = y^2$, $d(Y/X_0^2) = \eta(-df_1 + df_2 - df_3)/2$, $\eta \neq 0$, on both full orbits; translation $(ix_1, -ix_2, y)$ exchanges $[1:0:\pm 1]$, and the unique lift fixing the point retains full stabilizers. Put $V = \sum \nu_{1,c}, B = \sum \nu_{3,c}, h = -3j + B + R$; $V \geq B/2$ and $(b_1, b_2, b_3) = (vc_1, c_1 + vc_2, c_2 + vc_3)$. The degree formulas, with $A_c \leq 13$, give [15, Proposition 3.4] the tangent equality $2t = 2g - 2 + s + 3R/4 + \sigma$ and

$$a = \sum A_c + 15R/4 + \sigma - 26t \leq 26(1-g) - 6R - 12\sigma.$$

Consequently $C = \mathbb{P}^1$, $a = V + 2h + R + 2$, and $5h \leq 2a - 3j - R - 4 < 0$. The first residual degree satisfies $-2j + B - V \leq (a - 4j - 3R - 2)/5 < 0$. The row associated with the invariant of greatest degree and [15, Lemma 3.5] give $\dim \mathrm{Hom}(f^*T_X, \mathcal{O}_C(h)) = 1$; [15, Proposition 2.2] gives two coincident labels and $a \geq -3h$, hence $11a \geq 9j + 3R + 12$, a contradiction. □

Proposition 6.2. Every actual reflection in Γ has order 2. The square of $\mathcal{L}$ descends to an ample line bundle D on X. For every triple coincidence plane Z_I, write $\mathcal{L}|_{Z_I} \equiv \lambda_I \mathcal{O}_{Z_I}(1)$. Then $2\lambda_I$ is a positive integer, $\lambda_I \geq 1$, and $L - H$ is nef.

Proof. Meyer in rational dimension 8 gives a cusp of any actual order 4 mirror in the original lattice, with faithful horizontal reflection of order 4. The character fields [7, Table 1] and the trace $\mathrm{tr}(\mathrm{diag}(\zeta_m, \zeta_m^{-1}, 1)(23)) = \zeta_m$ leave $m = 1, 2, 4$ and the full horizontal list

$$A_3, \quad B_3, \quad G(4,4,3), \quad G(4,2,3), \quad G(4,1,3).$$

The last group alone has an order 4 reflection and is excluded by Lemma 6.1. The normalized reflections generate the lift $G \subset \mathrm{U}(h, K)$ of the same Γ; its determinant ± 1 forces every scalar ζI_5 to have $\zeta = \pm 1$, so [15, Proposition 2.11] descends $D = 2L$ with the full canonical frame at the cusp. For plane ranks $0, 1$, Lemma 3.2 gives $\lambda_I \geq 1, 2/3$; a quadratic minimum in rank 2 gives $2\lambda_I > 1$. For $G(4,2,2)$, the invariants $(v+S, vS+P^2, u^2P)$, $v = u^4$, give frame $(e_S, u^2 e_P)$, correction $-1/2$, and determinant order 2, hence $8\lambda_I = 2 + 3d_C/2$. The value $\lambda_I = 1/2$ would give $d_C = 4/3$, so half integrality gives $\lambda_I \geq 1$ for every plane; [15, Proposition 2.6] on all 35 planes gives $D - 2H$ nef. □

6.2 A lattice preserved by the original group

Proposition 6.3. After a Hermitian similarity over K, the same projective group Γ is a finite index subgroup of $\mathrm{SU}(I_{1,4}, \mathbb{Z}[i])$, where $I_{1,4} = \mathrm{diag}(-1, 1, 1, 1, 1)$. The similarity preserves the automorphic bundle, the full finite stabilizers, the full fixed normalizers, and the indices of all central cusp subgroups.

Proof. Let A be the Gram matrix, conjugate linear in the first variable, and G the original normalized reflection lift preserving a Gaussian lattice. Over K, irreducibility and Schur identify each full horizontal form with aI_3, $a > 0$ rational, with root norms $1, 2$; its determinant times a root norm is a norm because $a^4 = \mathrm{N}(a^2)$. Meyer gives a cusp of each actual mirror in dimension 8, and Witt decomposition gives

$$\det h = -\mathrm{N}(\alpha) \det h_{\mathrm{hor}}, \qquad h(r,r) = h_{\mathrm{hor}}(\bar{r}, \bar{r}).$$

Thus $(-\det A)h(r,r) \in \mathrm{N}(K^*)$; set $c = -\det A, h' = ch$, with $\det h' = -c^6$ and even valuations of root norms at odd inert primes. At odd inert p, maximize G-stable integral M; enlargement by $p^{-1}(M \cap p^2 M^\#)$ forces valuations $0, 1$, $pM^\# \subset M$, and even dimensional discriminant space. For a primitive root of norm n, integrality gives $h'(r, M) \subset n\mathbb{Z}_p[i]$, $r/n \in M^\#$; primitivity forces $v_p(n) \leq 1$, so its even parity gives $v_p(n) = 0$. Thus $(R_r - I)M^\# \subset M$, and the same G acts trivially on the discriminant space. A nonzero discriminant space has an isotropic line enlarging M, so $M = M^\#$. At 2, put $\pi = 1+i$ and maximize a G-stable integral lattice M. For a primitive reflection root $r \in M$ of norm n, the equality $R_r(M) = M$ gives $h'(r, M) \subset (n/2)\mathbb{Z}_2[i]$. If $v_2(n) \geq 2$, put $\Delta = \sum_{g \in G} \mathbb{Z}_2[i] gr$ and $\widetilde{M} = M + \pi^{-1}\Delta$. The inclusions

$$h'(M, \pi^{-1}\Delta) \subset \frac{n}{2\pi}\mathbb{Z}_2[i], \qquad h'(\pi^{-1}\Delta, \pi^{-1}\Delta) \subset \frac{n}{4}\mathbb{Z}_2[i]$$

show that $\widetilde{M}$ is integral and G-stable; primitivity of r makes it strictly larger than M, a contradiction. Hence $v_2(n) \leq 1$, so $(R_r - I)M^\# \subset M$. A maximal integral lattice $M' \supset M$ is self-dual: a vector v lifting a nonzero Gram radical has pairing in $\pi\mathbb{Z}_2[i]$ and norm in $2\mathbb{Z}_2$, so adjoining v/π would otherwise enlarge M'. Since $M \subset M' \subset M^\#$, every generating reflection satisfies $(R_r - I)M' \subset M \subset M'$; thus M' is stable under the original G. At split primes pair a stable module with its perfect dual; intersection patches these local self-dual modules globally, since only finitely many differ from the original lattice. A primitive isotropic vector splits

an integral plane $\left(\begin{smallmatrix} 0 & 1 \\ 1 & k \end{smallmatrix}\right)$; its positive self-dual complement is I_3, by successive unit vectors. If a positive self-dual Gaussian lattice of rank $r \le 3$ has no unit vector, its distinct points have squared distance at least 2. Any m points x_j of a translate in the unit ball then satisfy $m\sum|x_j|^2 - |\sum x_j|^2 \ge m(m-1)$, hence $\sum \max(1-|x_j|^2, 0) \le 1$. Integration over a fundamental domain of covolume one would give $\pi^r/((r+1)r!) \le 1$, which is false for $r = 1, 2, 3$. A unit vector splits off an orthogonal copy of $\mathbb{Z}[i]$, and the complement is again self-dual. Make k even with a unit vector, split a hyperbolic basis v, w, and use $(v-w+e, v+e, w-e)$ to obtain $I_{1,4}$. The basis $B^*(cA)B = I_{1,4}$ and lift $g \mapsto (\det g)^{-1}g$ put the original Γ in the special unitary group, with finite index. Here the normalized reflections have determinant -1, so every g in their generated group has $\det g = \pm 1$, and the determinant of its new lift is $(\det g)^{-4} = 1$. Uniqueness uses the absence of nontrivial fifth roots of unity in K. The equivariant tautological map and rescaling by c preserve every stated stabilizer, normalizer, bundle, and central index. □

6.3 Degrees on the coincidence planes

Lemma 6.4. The generic group of a coincidence plane of rank 2 is A_1^2, A_2, B_2, or $G(4, 2, 2)$. An A_2 plane never occurs. A B_2 plane has $\lambda > 1$.

Proof. Meyer in rational dimension 6 embeds the full generic group faithfully into a horizontal parabolic group; its orders $4, 6, 8, 16$ exclude $I_2(6)$, and the character field and [10, Theorem 3.6] give the list. For A_2, the axes in $A_3, G(4, 4, 3), G(4, 2, 3)$ are $(u^2, u^3, u^4), (u^3, 3u^4, 3u^8), (3v^2, v^3, 3v^4)$, $v = u^2$; their zero differentials contradict smooth embedding. The remaining B_3 has matrix $\left(\begin{smallmatrix} 4u^2 & 4u^2 \\ 12u^4 & 20u^4 \end{smallmatrix}\right)$, giving $d_B = 2\lambda - 4/5$, contrary to half integrality. For B_2, put $q_2 = xy, q_4 = x^4 + y^4$; the invariants $(uq_2, u^4 + q_4, u^4q_4 + q_2^4)$ give matrix $\mathrm{diag}(v, v)$, $v = u^4$, and correction $-3/4$; B_3 gives $\left(\begin{smallmatrix} v & 1 \\ 0 & v \end{smallmatrix}\right)$, $v = u^2$. Hence $6\lambda - 2 = 5d_C/4 + 2d_B$, whose sole integral solution at $\lambda = 1$ is $(0, 2)$. The degree 4 direction is preserved by the full normalizer, has trivial divisorial character, and its residual finite character is trivial on finitely punctured $\mathbb{P}^2$; its row of the inverse differential associated with the invariant of greatest degree gives $\mathcal{O}_Z(-1)^2 \to \mathcal{O}_Z(-4\lambda + d_C + d_B) = \mathcal{O}_Z(-2)$, a contradiction. □

Lemma 6.5. A coincidence plane with generic group A_1^2 has $\lambda > 1$.

Proof. Let $d_A, d_B, d_C, d_P, d_Q, d_R$ be the degrees of the reduced curves with full groups

$$A_3,\ B_3,\ G(4, 2, 3),\ B_2 \times A_1,\ G(4, 2, 2) \times A_1,\ A_1^3.$$

The upper group $A_2 \times A_1$ has singular axis image and is excluded. The regular normal determinant and effective canonical branch formulas are

$$4\lambda - 2 = \frac{3}{2}d_A + 3d_B + \frac{5}{2}d_C + d_P + \frac{1}{2}d_Q,$$
$$3\lambda + 3 = \frac{1}{2}(d_A + d_B + d_P + d_R) + \frac{3}{4}(d_C + d_Q).$$

The local inputs and the full cusp restrictions are proved in Lemma A.1. At $\lambda = 1$, the integral possibilities are

$$(d_P, d_Q, d_R) = (0, 4, 6), (1, 2, 8), (2, 0, 10), \qquad d_A = d_B = d_C = 0.$$

The formal row $d_A = d_Q = 1$ gives $d_R = 19/2$. For row one, count the three finite germs of Lemma A.1 by A, B, C_0, cusp nodes and tacnodes by N_0, T_0; for $x = \sum(g_j - 1)$, the full normalizers give

$$2x + 2(N_0 + T_0) + A/2 + 3B/4 + 3C_0/2 = 8,$$
$$x = 2 - C_0 - N_0 - 2T_0,$$
$$A + 2B + 2N_0 = 24.$$

Thus $20 \le 3\delta + 2T_0$, $\delta = C_0 + N_0 + 2T_0$; a reduced quartic has $\delta \le r + 2$, and degree partitions bound tacnodes by $1, 1, 2, 2, 0$, making the right side at most 19. In rows two and three, the full cusp data are $(|F_\Pi|, \deg M_\Pi) = (4, 2 \text{ or } 4), (2, 2)$, each with one cusp. Euler characteristic 3 gives maximal index 96 and $(|F_\Delta|, \deg M_\Delta) = (4, 2 \text{ or } 4)$ by Lemma A.2; curvature and the unique cusp double coset give

$$[\Delta : \Pi] = \frac{|F_\Delta|}{|F_\Pi|} \frac{\deg M_\Pi}{\deg M_\Delta} h^2.$$

The four noncentral quantities are powers of 2, so the index has even 3-adic valuation, contradicting 96. □

6.4 The fixed surface with group $G(4, 2, 2)$

Put $T_2 = G(4, 2, 2)$ and $T_3 = G(4, 2, 3)$. Suppose that a coincidence plane $Z = Z_I$ has full generic group T_2. [15, Lemma 2.12] identifies the compactification of its full fixed ball quotient with Z. Generic identifications normalize T_2, the map is finite, and Z is normal. Write D_C, D_P for the reduced curves with generic full groups $T_3, T_2 \times A_1$, of effective branch orders $4, 2$.

Lemma 6.6. The special interior groups on D_C and their full fixed surface germs are

	D_C	D_P
$G(4,2,4)$	$y = 0$	$x^2 - 4y = 0$
G_{31}	$y^2 - x^3 = 0$	$\varnothing$
$T_3 \times A_1$	$x = 0$	$y = 0.$

At G_{31}, the image on the two indecomposable invariants of degree 4 is the full S_3. The full cusp germs and the degrees of their two surface coordinates are

(d_1, d_2, d_3)	actual degrees	$\deg(B, C)$	D_C	D_P
211	1224	$(1, 2)$	$C^2 + B^4$	$\varnothing$
221	1112	$(1, 1)$	$C^2 + B^2$	B
212	1244	$(1, 4)$	$C + B^4$	C
222	1122	$(1, 2)$	$C + B^2$	$BC.$

Proof. The full parabolic normalizers [10, Theorems 2.1, 3.6, 5.14 and Tables 5, 6, 8] give the first table: use $(z_1^4 + z_2^4, z_1^4 z_2^4)$ for $G(4, 2, 4)$, independent axes for products, and for G_{31}

$$f_8 = a^8 + 14a^4b^4 + b^8, \qquad f_{12} = a^{12} - 33a^8b^4 - 33a^4b^8 + b^{12},$$
$$J = -1728a^3b^3(a^4 - b^4)^3, \qquad f_8^3 - f_{12}^2 = 108a^4b^4(a^4 - b^4)^4.$$

The six mirrors $0, \infty, \pm 1, \pm i$ give the full G_8 of order 96; on $(S, P) = (a^4 + b^4, (ab)^2)$, generators $A = (1 - i)^{-1} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), B = \mathrm{diag}(i, 1)$ act by $(-S/2 - 3P, -S/4 + P/2), (S, -P)$, giving S_3. For the full torsor [15, Lemma 2.17], take all permutations and phase products in $\mu_{d_1} \times \mu_{d_2} \times \mu_{d_3} \subset \mu_4(x_1) \times \mu_4(x_2) \times \mu_2(y)$, $d_j \in \{1, 2\}$. Every component and affine translate is (a, a, t), $a = (0, 1, i), d_1 = 2$; the other branch supports give A_1^2, B_2 or independent squares, and 111 gives none. The sections restrict to $(0, q^{d_2} x_2^{d_2}, (-1)^{d_3} q^{2d_3} y^{d_3}, q^4 x_1^4)$, yielding the four polynomial rings and divisors after removing exactly $g = d_2$, $(q, c) \mapsto (q^g, c/g)$. The unique lift fixing the point preserves full stabilizers, and every pointwise scalar is trivial on q^g; only further quotients $1224, 1244 \mapsto 1112, 1122$ are smooth, so no component or smaller period is omitted. □

Lemma 6.7. There is a finite flat normal cover $p : Y \to Z$ of degree 3, whose trace kernel is

$$\mathcal{B} = N_{Z/X}(-D_C) \otimes \mathcal{L}^4|_Z = \mathcal{O}_Z(-b)^{\oplus 2}.$$

Its parameters satisfy

$$b = (4\lambda - 1)/3, \qquad \deg D_C = 4b, \qquad \deg D_P = 3(5-b)/2, \qquad b \in \{1, 3, 5\}.$$

There are exactly $3b^2$ distinct G_{31} points.

Proof. Let Π_Z be the effective full normalizer of the fixed ball, and let $\Sigma \subset Z$ contain the cusps, special interior points, and singular points of $D_C \cup D_P$. The image on the positive definite normal lattice is finite. The projective image of T_2 is the Klein four group, whose three nonidentity elements have fixed point pairs $\mathcal{P} = \{\{0, \infty\}, \{1, -1\}, \{i, -i\}\}$. Its normalizer in $\mathrm{PGL}(2, \mathbb{C})$ permutes these three pairs; the kernel is the projective image of T_2, and the preceding matrices A, B give two transpositions. Thus every normal block has the form $Q = cQ_0$, with $Q_0 \in \langle T_2, A, B\rangle$ and $c \in K^*$. Both matrices have finite order, so $c \in \mu_4$. The scalar c acts trivially on $(S, P) = (y_1^4 + y_2^4, (y_1 y_2)^2)$. The pointwise group T_2 and the central lifts also act trivially. The resulting homomorphism $\rho : \Pi_Z \to S_3$ acts by the standard representation, which is self-dual through the invariant form $S^2 + 12P^2$.

Put $U = Z \setminus (D_C \cup \Sigma)$. The inertia along D_P is trivial under ρ, so the permutation action on the three pairs defines a finite étale cover of degree 3 over U. Explicitly, if $\mathbb{B}_2^\circ$ is the inverse image of U, this cover is $(\mathbb{B}_2^\circ \times \mathcal{P})/\Pi_Z \to U$, with $\gamma(z, p) = (\gamma z, \rho(\gamma)p)$ and $[(z, p)] \mapsto [z]$. Normalize Z in the function fields of its connected components to obtain $p : Y \to Z$; this is the associated cover of degree 3, rather than the principal cover defined by $\ker \rho$. We obtain flatness from normality in dimension 2 and the Auslander–Buchsbaum formula, and trace splits $p_*\mathcal{O}_Y = \mathcal{O}_Z \oplus \mathcal{B}$.

If $j(\gamma, z)$ is the automorphy factor on the fixed ball, the normal coordinates transform by $y \mapsto Qy/j(\gamma, z)$. Hence the homogeneous normal directions transform by $\rho(\gamma)j(\gamma, z)^{-4}$, and $\mathcal{L}^4$ removes precisely this automorphy factor. The bundle $\mathcal{L}^4$ descends on all of Z because $2\mathcal{L}$ already descends on X. At a general point of D_C, put $v = u^4$. The local cover is the disjoint union of one unramified sheet and $s^2 = v$, where $s = u^2$, so the trace kernel has frames $(2, -1, -1), (0, u^2, -u^2)$. These frames identify with the invariant normal vectors $(e_S, u^2 e_P)$. After the tangent direction is removed from the invariants $(v + S, vS + P^2, u^2 P)$, their normal differential is vI_2, with image $N_{Z/X}(-D_C)$. At D_P, the cover is unramified and the normal differential is invertible. We obtain $\mathcal{B} \otimes \mathcal{L}^{-4} \simeq N_{Z/X}(-D_C)$ on $Z \setminus \Sigma$, including the regular lattices along every divisor. Normalization uniquely determines the cover at the special points and cusps; local freeness on both sides then extends the bundle isomorphism across Σ by Hartogs. The trace discriminant and the fixed ball canonical formula give $d_C = 4(d_C + 1 - 4\lambda)$ and $-3 + 3d_C/4 + d_P/2 = 3\lambda$. Miranda's cubic $F = a_0 s^3 + 3a_1 s^2 t + 3a_2 s t^2 + a_3 t^3$, $a_i \in H^0(\mathcal{O}_Z(b))$, defines finite $\phi : Z \to \mathbb{P}^3$, since a common zero would give multiplicity 4 [13, Theorem 3.6]. The discriminant germ $v^2 - u^3$ along the twisted cubic W forces simple inverse points by the permitted local types; conversely, the minors of $\left(\begin{smallmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_3 \end{smallmatrix}\right)$ characterize every G_{31} point [13, Corollary 4.6]. The Hilbert–Burch sequence

$$0 \to \mathcal{O}_Z(-3b)^2 \to \mathcal{O}_Z(-2b)^3 \to \mathcal{O}_Z \to \mathcal{O}_{\phi^{-1}(W)} \to 0$$

gives total length $3b^2$. □

Lemma 6.8. For a smooth curve C in a fixed ball surface with $2L$ descended,

$$2L \cdot C \le 2g(C) - 2 + |S| + \sum_{z \notin S} (1 - 1/e_z).$$

Here S is its cusp set, and e_z is the effective inertia on the normalization of a lifted branch. Equality makes every lifted branch a complex geodesic.

Proof. On a normal neat cover of the same linear group, kill finite characters, normalize every inverse component, and divide Riemann–Hurwitz by the degree after removing the pointwise kernel. The restricted ambient canonical frame at the cusp introduces no parameter power; for curvature -1, $c_1(2L) = \omega/(2\pi)$, and the potential $-2\log(-c\log|t| + A(t))$ for $q = t^m a(t)$ has zero Lelong number and Stokes limit. Schwarz and Gauss–Bonnet give the formula, and metric equality gives a complex geodesic. □

Proposition 6.9. A coincidence plane with full generic group $G(4,2,2)$ has $\lambda > 1$.

Proof. Suppose $\lambda = 1$; then $b = (4\lambda - 1)/3 = 1$ by Lemma 6.7. For $b = 1$, the plane coefficient map meets the twisted cubic in three simple points; after $\mathrm{PGL}(2,\mathbb{C})$, its cubic, discriminant normalization, and tangent map are

$$F = x(s^3 - t^3) + 3ys^2t + 3zst^2,$$
$$[3s^2t^2 : -t(2s^3 + t^3) : s(s^3 + 2t^3)], \quad [s^3 - t^3 : 3s^2t : 3st^2].$$

The syzygies $(0, t, -s), (3st, -s^2, t^2)$ give root incidence surface $\mathbb{F}_1$; the tangent map identifies only $[1:0], [0:1]$, so D_C is tricuspidal with unique bitangent $\ell = (x = 0)$. For intersection k with the reduced mirror divisor, omitting a component equal to the curve, effective inertia satisfies $1 - 1/e \le k/2$: use $2\operatorname{ord} J = ek$, $\operatorname{ord} J \ge e - 1 + (\nu - 1)^2$; for a mirror component use $2(e - 1 + \operatorname{ord} g) = \operatorname{ord} g + ek$. Equality requires one transverse reflection, or two orthogonal reflections for a component. If $\ell \not\subset D_P$, its tangencies contribute $1/2$ each, strictly less than $1/2 + (\ell \cdot D_P)_z/2$ at special finite points or cusps; the full cusp table gives intersection at least 2, and all cusps lie on D_C. Since $d_P = 6$, Schwarz gives equality in $2 = 2L \cdot \ell \le -2 + |S| + \sum(1 - 1/e_z) \le -2 + 1 + 3 = 2$. Both general interior branches $v = \pm u^2$ would be geodesic with the same point and tangent, contradicting uniqueness. If $\ell \subset D_P$, the bound is again $-2 + 2(3/4) + 5/2 = 2$, with equality forcing interior $G(4,2,4)$ tangencies and no cusp. Every D_P component nevertheless has a cusp of the same Γ: its T_2 support has form aI_2 and norm determinant, and self-duality gives primitive root norm 1 or 2, so the negative mirror determinant is $\mathrm{N}(\delta)$. For orthogonal norms $\beta > 0, \gamma < 0$, $-\beta\gamma = \mathrm{N}(\delta)$ gives isotropic $w = (\delta/\beta, 1)$; $Q(x) = h(w,x)w$ satisfies $Q^2 = 0$, and a positive power of the integral $I + miQ$ belongs to the original Γ and centralizes T_2. □

Proposition 6.10. For every triple coincidence plane, $\lambda_I \ge 3/2$. In particular, $2L - 3H$ is nef.

Proof. Lemma 3.3, Lemmas 6.4, 6.5, and Proposition 6.9 exclude $\lambda_I = 1$ in every rank. Half integrality from Proposition 6.2 gives $\lambda_I \ge 3/2$; apply [15, Proposition 2.6] to $D - 3H$ on each of the 35 planes. □

7 Chern classes and the seven point quotient

7.1 Chern classes of all fixed strata

Let X be a smooth projective variety of dimension d isomorphic to the Baily–Borel compactification of an arithmetic ball quotient. We keep its lattice Γ and automorphic class L fixed. For each full stabilizer P, let V_P be its nontrivial reflection representation and let Z_P be the sum of the reduced closures of the strata with generic stabilizer P. Count each component once on its full normalizer quotient. Put $\lambda_P = \#\{g \in P : V_P^g = 0\}/|P|$, $\mathcal{C}_k = \sum_{\operatorname{rank} P = k} \lambda_P [Z_P]$, and $\mathcal{C}_0 = [X]$. If $e_{P,1}, \dots, e_{P,k}$ are the degrees of the basic invariants, then

$$\lambda_P = \frac{\prod_i (e_{P,i} - 1)}{\prod_i e_{P,i}}. \tag{7.1}$$

The fixed space polynomial is $\sum_{g\in P} t^{\dim V_P^g} = \prod_i (t + e_{P,i} - 1)$; see [23, 7].

Proposition 7.1. Assume that the quotient of every positive dimensional fixed ball by its full normalizer has finite volume and has automorphic class equal to the restriction of L. For $0 \le k < d$, the following equality holds after pairing with products of divisor classes:

$$\mathcal{C}_k = \sum_{j=0}^{k} \binom{d-k+j+1}{j} L^j c_{k-j}(T_X). \tag{7.2}$$

It is an equality of numerical cycle classes if these pairings detect numerical equivalence. If c is the number of cusps, then

$$c + \deg \mathcal{C}_d = \int_X \sum_{j=0}^{d} (j+1) L^j c_{d-j}(T_X). \tag{7.3}$$

Proof. On a normal neat cover of the same Γ, apply Burnside's formula to fixed balls. For a full generic stabilizer P, normalizer $\mathcal{N}$, and centralizer $\mathcal{Z}$ of an element fixing precisely this ball, the coefficient on the effective quotient is $[\mathcal{N} : \mathcal{Z}P]/|\mathcal{Z} \cap P| = [\mathcal{N} : \mathcal{Z}]/|P|$. Summing its conjugacy classes gives λ_P, with every local branch retained. For $k < d$, a general complete intersection of $d - k$ sufficiently ample divisors avoids the cusps and higher rank strata. The fixed ball of dimension $d - s$ has orbifold Chern polynomial $(1 - Lt)^{d-s+1}$ by [15, Equation 3.13]. Write $A_1, \dots, A_{d-k}$ for the divisor classes of this complete intersection Y, and put $A = A_1 \cdots A_{d-k}$. Burnside's formula and Gauss–Bonnet on each normalized fixed component give

$$e(Y) = \sum_{s=0}^{k} \int_X A\mathcal{C}_s \left[\frac{(1-L)^{d-s+1}}{\prod_{i=1}^{d-k}(1+A_i)} \right]_{k-s},$$

where brackets denote the component of the indicated codimension. On the other hand, adjunction gives $e(Y) = \int_X A[c(T_X)/\prod_i (1 + A_i)]_k$. Canceling the common denominator inductively in k, and polarizing the ample divisor classes, gives

$$c_k(T_X) = \sum_{s=0}^{k} (-1)^{k-s} \binom{d-s+1}{k-s} L^{k-s} \mathcal{C}_s.$$

Binomial inversion proves (7.2). For $k = d$, use compactly supported Euler characteristics on the open quotient and finite volume Gauss–Bonnet; the left side of the triangular equation becomes $\int_X c_d(T_X) - c$. Inverting gives (7.3); ends of fixed components are already in the open average. □

[15, Lemma 2.12] supplies the finite volume and full normalizer hypotheses below.

7.2 All Gaussian fixed components

Retain the same group and integral lattice of Proposition 6.3, and put $H = -K_X$, $c_j = c_j(T_X)$. Every reflection has order 2, and its primitive root has norm 1 or 2. Proposition 6.10 gives

$$N = L - \frac{3}{2} H \text{ nef}. \tag{7.4}$$

We do not exclude either root norm or the reflection subgroups $G(4,2,2)$ and $G(4,2,3)$.

For a reflection group G, put $d_G = \sum_i e_{G,i}$, where the $e_{G,i}$ are its basic invariant degrees. The generic groups on fixed surfaces are $P_D = A_1^2$, $P_A = A_2$, $P_B = B_2$, and $P_T = G(4,2,2)$.

Their degrees, weights from (7.1), and ordered mirror intersection coefficients are

$$\begin{array}{c|c|c|c} P & (e_{P,i}) & \lambda_P & c_P \\ \hline P_D & (2,2) & 1/4 & 2 \\ P_A & (2,3) & 1/3 & 4 \\ P_B & (2,4) & 3/8 & 6 \\ P_T & (4,4) & 9/16 & 15/2 \end{array}, \qquad c_P = \frac{4\nu_P(\nu_P-1)}{|P|}, \quad \nu_P = d_P - 2.$$

The rank 2 classification over $\mathbb{Q}(i)$ leaves these groups and $I_2(6)$ among groups generated by reflections of order 2 [7]. For $I_2(6)$, primitive root norms $a, b \in \{1,2\}$ would satisfy $4|h(r,s)|^2 = 3ab$. The only integral possibility requires a Gaussian integer of norm 3, which does not exist. The groups on fixed curves are listed in Table 1. This list follows directly from the rank 3 classification over $\mathbb{Q}(i)$; no cusp is assumed on an individual curve.

K_k	$(e_{K_k,i})$	$\lvert K_k\rvert$	$(n_{D,K_k}, n_{A,K_k}, n_{B,K_k}, n_{T,K_k})$	λ_{K_k}
$K_0 = A_1^3$	$(2,2,2)$	8	$(3,0,0,0)$	$1/8$
$K_1 = A_2 \times A_1$	$(2,2,3)$	12	$(3,1,0,0)$	$1/6$
$K_2 = B_2 \times A_1$	$(2,2,4)$	16	$(4,0,1,0)$	$3/16$
$K_3 = A_3$	$(2,3,4)$	24	$(3,4,0,0)$	$1/4$
$K_4 = B_3$	$(2,4,6)$	48	$(6,4,3,0)$	$5/16$
$K_5 = G(4,4,3)$	$(3,4,8)$	96	$(0,16,3,0)$	$7/16$
$K_6 = G(4,2,2) \times A_1$	$(2,4,4)$	32	$(6,0,0,1)$	$9/32$
$K_7 = G(4,2,3)$	$(4,6,8)$	192	$(12,16,0,3)$	$35/64$

Table 1: Groups on fixed curves and their parabolic supports.

Write Z_P for the sum of the reduced fixed surfaces of generic type P, and W_k for the corresponding sum of curves of type K_k in the table. Each component is counted once on its full normalizer quotient; different local branches of a self-intersection remain distinct. The number $n_{P,K}$ counts subspaces with full pointwise stabilizer P in K. Products give the reducible rows in the table. For $G(4,2,3)$, the coordinate and phase equality axes give $12, 16, 0, 3$ supports of the four surface types; for $G(4,4,3)$, they give $0, 16, 3, 0$. Define

$$v_{P,K} = (d_K - d_P - 1)\frac{|P| n_{P,K}}{|K|}.$$

No rank 4 parabolic counts will be needed.

Lemma 7.2. With the classes ψ_i, p of Lemma 2.1, one has

$$\begin{gathered} \int_X H^4 = 154, \qquad \int_X H^2 p = -14, \qquad \int_X p^2 = 10, \qquad \int_X c_4 = 38, \\ Hp \equiv -\frac{1}{11}H^3, \qquad c_3 \equiv \frac{4}{11}H^3, \qquad p^2 \equiv \frac{5}{77}H^4. \end{gathered} \tag{7.5}$$

The nef cone is generated by $U_i = (H - \psi_i)/2$ and $V_i = (H + \psi_i)/2$, $1 \le i \le 7$. For $E = \frac{1}{2}\sum_i a_i\psi_i$, $E' = \frac{1}{2}\sum_i a_i'\psi_i$, $s = \sum_i a_i$, and $s' = \sum_i a_i'$, the intersection form is

$$\int_X H^2 EE' = \frac{3}{2}ss' - 5\sum_i a_i a_i', \qquad -\int_X pEE' = \frac{1}{2}ss' - 3\sum_i a_i a_i'. \tag{7.6}$$

Proof. For a chamber divisor D of multidegree (b_i), put $q = \frac{1}{2}\sum b_i$ and $d_I = q - \sum_{i\in I} b_i$. Subtracting torus weight multiplicities 2 from 0 in the product of seven characters gives

$$h^0(X, kD) = \sum_{|I|\le 3} (-1)^{|I|} \binom{kd_I + 5 - |I|}{5}.$$

Its leading coefficient yields $\int_X D^4 = 3q^4 - 2\sum_i (q-b_i)^4 + \sum_{i<j}(q-b_i-b_j)^4$. Polarize and use $\psi_i^2 = p$: the intersections of four distinct ψ_i are 2, $p\psi_i\psi_j = -2$ for $i \neq j$, and $p^2 = 10$. Expanding $H = \sum_i \psi_i$ and the tangent sequence gives the stated numbers, (7.6), and $c_4 = (H^4 - 22H^2p + 45p^2)/24$. Divisor classes generate the rational Chow ring [5, Lemma 8, Theorem 10, Remark 3], so the symmetric pairings determine the numerical relations. The criterion of [15, Proposition 2.6], for triples of labels, gives the 14 nef generators. □

Let $\mathscr{B}$ be the reduced reflection divisor. Proposition 7.1 gives

$$\begin{gathered} \mathscr{B} = 10L + 2H, \qquad \sum_P \lambda_P Z_P = c_2 + 4LH + 10L^2, \\ \sum_{k=0}^{7} \lambda_{K_k} W_k = c_3 + 3Lc_2 + 6HL^2 + 10L^3. \end{gathered} \tag{7.7}$$

The normal class of a lifted mirror is $-L$. Counting ordered pairs of distinct local mirrors, including branches of one global divisor, gives $\sum_P c_P Z_P = \mathscr{B}^2 + 2L\mathscr{B}$. Solving the two surface equations yields

$$\begin{aligned} Z_D + \frac{2}{3}Z_A + \frac{3}{4}Z_T &= 20L^2 + 10LH + 2H^2 - 20p, \\ \frac{2}{3}Z_A + \frac{3}{2}Z_B + \frac{3}{2}Z_T &= 20L^2 + 6LH + 10p. \end{aligned} \tag{7.8}$$

Proposition 3.1 of [15], with $d = 4, k = m = 2$, gives

$$(H + (d_P + 3)L)Z_P = \sum_k v_{P,K_k} W_k. \tag{7.9}$$

Finally, if c is the number of cusps and n_J the number of points of full rank 4 type J, put

$$F(L) = \int_X (c_4 + 2Lc_3 + 3L^2c_2 + 4HL^3 + 5L^4) = c + \sum_J \lambda_J n_J. \tag{7.10}$$

7.3 The same determinant on every curve

For an interior group J, let $\gamma_k(J) = |K_k| n_{K_k,J}/|J|$, where $n_{K,J}$ counts its axes with full pointwise group K. At a cusp c', let $\gamma_k(c')$ be the degree of the stack points with nonzero fiber stabilizer K_k in the weighted projective quotient of the horizontal abelian threefold. Write $b(c')$ for the generator degrees measured by the full ambient central period, and put $s_{c'} = \sum_j b_j(c')$. With $A_k = \int_X HW_k$, $T_k = \int_X LW_k$, Proposition 3.1 of [15] gives

$$A_k + (d_{K_k} + 2)T_k = \sum_J (d_J - d_{K_k} - 1)\gamma_k(J) n_J + \sum_{c' \text{ cusp}} s_{c'}\gamma_k(c'). \tag{7.11}$$

Every local axis extends to a fixed curve for the full normalizer by [15, Lemma 2.12]. The sum over its local axis orbits is $\sum 1/e = |K| n_{K,J}/|J|$. At a cusp, [15, Lemma 2.17] gives the unique lift fixing the point in the same Γ; its disk supplies the corresponding curve end, with contribution $1/\kappa$ for the full fiber character of order κ. Thus every stack point counted by $\gamma_k(c')$ occurs, with its actual width. The identity includes compact fixed curves, for which the cusp sum is empty. Nefness in (7.4) gives $T_k \geq 3A_k/2$, independently of the degree and genus of each component.

The possible interior basic degree tuples are

$$\begin{gathered} (2,2,2,2), \quad (2,2,2,3), \quad (2,2,2,4), \quad (2,2,3,3), \\ (2,2,3,4), \quad (2,2,4,4), \quad (2,2,4,6), \quad (2,3,4,4), \\ (2,3,4,5), \quad (2,3,4,8), \quad (2,4,4,4), \quad (2,4,4,6), \\ (2,4,6,8), \quad (2,6,8,12), \quad (4,4,4,4), \quad (4,4,8,12), \\ (4,8,8,12), \quad (4,8,12,20), \quad (8,12,20,24). \end{gathered} \tag{7.12}$$

The irreducible groups are $A_4, D_4, B_4, F_4, G(4,4,4), G(4,2,4), G_{29}, G_{31}$; the remaining tuples come from products [23, 7]. At F_4, the only rank 3 parabolic types are $A_2 \times A_1$ and B_3, by deleting one vertex of its Coxeter diagram. Accordingly $\gamma_k(F_4) = 0$ for $k \notin \{1, 4\}$. No other interior support restriction is used.

The full cusp degrees lie in

$$1111, \quad 1112, \quad 1122, \quad 1222, \quad 1124, \quad 1144, \quad 1224, \quad 1244. \tag{7.13}$$

The full cusp group is generated by actual reflections, so its horizontal image is generated by reflections of order 2. The horizontal types are $A_3, B_3, G(4,4,3), G(4,2,3)$; the phase product character excludes $G(4,1,3)$ from such a generating set. For $G(4,2,3)$, the five rows of [21, Section 4.4], including its torsor, have parameters

$$(d_1, d_2, d_3) = (1,1,2), (1,2,1), (1,2,2), (2,2,1), (2,2,2).$$

The model in [21, Example 3.10] has degrees $(d_1, d_2, 2d_3, 4)$. Removing exactly the scalar fiber kernel gives $1144, 1224, 1244, 1112, 1122$, respectively; $G(4,4,3)$ gives 1124, and A_3, B_3 give the first four tuples. This operation is $q_{\mathrm{new}} = q_{\mathrm{old}}^g$, $c_{\mathrm{new}} = c_{\mathrm{old}}/g$, for the scalar order g, and returns the grading to the actual full central period. Descent along a free translation cover retains the linearization of the full group and the unique lifts fixing the points. Any further smooth scalar quotient has order dividing $\operatorname{lcm}_j \gcd_{i \neq j} b_i$, since the finite quotient group is generated by reflections. The only nontrivial cases are $1222 \to 1111$, $1224 \to 1112$, and $1244 \to 1122$, so no additional tuple occurs.

7.4 Finite quotients with the same incidence equations

Proposition 7.3. Let $M = \mathbb{P}^3$, $\Theta = \mathcal{O}(1)$, $\varepsilon = 1$, or let M be an abelian threefold with an ample line bundle Θ, $\varepsilon = 0$. Let a finite group act on (M, Θ), faithfully on the punctured total space of Θ^{-1}, with polynomial ring of invariant sections of degrees $b = (b_1, \dots, b_4)$. Assume that the punctured quotient is smooth, that every actual reflection has order 2, and that the rank 2 and rank 3 stabilizers have the types above. On the weighted projective stack of weights b, let β_P be the degree of the reduced curve images of generic type P, measured by $\eta = c_1(\mathcal{O}(1))$, and let γ_k be the degree of its stack points of type K_k. The group at a point is the kernel of the full character on the nonzero fiber, and all residual stabilizer orders are retained in these degrees. Put $s = \sum b_i$, $v = \prod b_i$, and let e_j denote the elementary symmetric polynomial of degree j. Then $\beta_P, \gamma_k \geq 0$ and

$$\begin{aligned} \sum_P \lambda_P \beta_P &= \frac{e_2(b-\varepsilon)}{v}, \qquad \sum_P c_P \beta_P = \frac{4(s-4\varepsilon)(s-5\varepsilon)}{v}, \\ \sum_k \lambda_{K_k} \gamma_k &= \frac{e_3(b-\varepsilon)}{v}, \qquad (s - \varepsilon(d_P+2))\beta_P = \sum_k v_{P,K_k} \gamma_k. \end{aligned} \tag{7.14}$$

Here $b - \varepsilon = (b_i - \varepsilon)_i$; any common divisor of the b_i is retained.

Proof. The sections give a finite map from M to the weighted projective stack, pulling $\mathcal{O}(1)$ back to Θ. The sections have no common zero on M; after base change by the atlas $\mathbb{A}^4 \setminus \{0\}$, this map is the finite quotient of the punctured Θ^{-1} torsor by the stated group. For a horizontal point with stabilizer F_x and fiber character χ_x, its residual order is $r = |\operatorname{im} \chi_x|$, and orbit counting assigns weight $|\ker \chi_x|/|F_x| = 1/r$. Thus a point contributes $1/r$ to γ_k; a curve contributes its degree against η, with the same weight at its generic point, to β_P. For example, a weighted projective coordinate point of residual order r has stack degree $1/r$, rather than coarse degree one. Apply the weighted Burnside count in Proposition 7.1 to general complete intersections. The fixed components are projective linear spaces for $\varepsilon = 1$ and translates

of subtori for $\varepsilon = 0$, with Chern polynomials $(1+\varepsilon\eta)^{4-a}$ in codimension a. The weighted Euler sequence gives $c_j = e_j(b)\eta^j$ and $\int \eta^3 = 1/v$. Binomial inversion yields the first and third equations. For the reduced branch divisor, ramification gives $\mathscr{D} = 2(s-4\varepsilon)\eta$. Counting distinct ordered mirrors, whose normal class is $\varepsilon\eta$, gives $\sum_P c_P \mathscr{Z}_P = \mathscr{D}^2 - 2\varepsilon\eta\mathscr{D}$, proving the second equation. On a normalized curve of type P, the tangent and the two invariant normal directions have total first Chern class $\varepsilon(d_P+2)\eta$. The same determinant calculation as in (7.9), with its full characters, gives the last four equations. □

7.5 A comparison of all components

For any two nef classes E, E', the fourteen generators in Lemma 7.2 give

$$0 \le -\int_X pEE' \le \frac{1}{7}\int_X H^2EE'. \tag{7.15}$$

Put $l = \int_X L^4 > 0$ and normalize all intersection numbers by l:

$$\begin{aligned}(z_P, h_P, j_P) &= l^{-1}\int_X (L^2Z_P, HLZ_P, H^2Z_P),\\ (a_k, w_k) &= l^{-1}(A_k, T_k),\\ (\rho, x, u, y, z_4) &= l^{-1}\int_X (HL^3, H^2L^2, -pL^2, H^3L, H^4).\end{aligned}$$

All these numbers are nonnegative. Nefness of N, L, H and (7.15) imply

$$\rho, x, y, z_4 \le 1, \qquad u \le 1/7, \qquad z_P, h_P, j_P \le 64, \qquad a_k, w_k \le 160, \qquad F(L)/l < 13. \tag{7.16}$$

One has $\rho \le 2/3$, $x \le 2\rho/3$, $y \le 2x/3$, and $z_4 \le 2y/3$. The two equations in (7.8), paired with L^2, bound their positive left sides by $35, 26$, respectively, and give $z_P < 64$. The inequalities $z_P \ge 3h_P/2 \ge 9j_P/4$ give the other surface bounds. The curve equation in (7.7), paired with L, gives

$$\sum_k \lambda_{K_k} w_k = 10 + 6\rho + \frac{3}{2}x + \frac{15}{2}u + \frac{4}{11}y < 19.$$

Since $\lambda_{K_k} \ge 1/8$ and $w_k \ge 3a_k/2$, the curve bounds follow. Finally,

$$F_0 = F(L)/l = 5 + 4\rho + \frac{3}{2}x + \frac{15}{2}u + \frac{8}{11}y + \frac{19}{77}z_4 < 13.$$

Lemma 7.4. Equations (7.7)–(7.11) and the local relations (7.14) contradict (7.4).

Proof. Use the actual local degree vectors (β_P, γ_k) from (7.14). Let $n_\nu, c_\nu \ge 0$ count the actual interior and cusp models, respectively, divided by l; the index includes their local degree vectors. For an interior tuple put $\Lambda_\nu = \prod_i (b_i - 1)/\prod_i b_i$. Then $F_0 = \sum_\nu \Lambda_\nu n_\nu + \sum_\nu c_\nu$.

Let R_1, R_2 be the left sides minus the right sides of (7.8). For $(D_1, D_2, D_3) = (L^2, HL, H^2)$, put $E_{2j-1} = l^{-1}\int_X R_1 D_j$ and $E_{2j} = l^{-1}\int_X R_2 D_j$. The next four relations, $E_7, \dots, E_{10}$, are (7.9) paired with L, in the order D, A, B, T. Let E_{11} be the curve equation in (7.7) paired with L, and let $E_{12} = F_0 - \sum \Lambda_\nu n_\nu - \sum c_\nu$. Let $E_{13}, \dots, E_{20}$ be (7.11), in curve order, with the actual local degrees and their normalized counts. Finally, $E_{21}, \dots, E_{24}$ are (7.9) paired with H, and E_{25} is the curve equation in (7.7) paired with H. All are left sides minus right sides, divided by l, and all vanish.

The following nine quantities are nonnegative:

$$\begin{gathered}G_P = h_P - \tfrac{3}{2}j_P \quad (P = D, A, B, T),\\ G_5 = \tfrac{1}{7}x - u - \tfrac{6}{77}y, \qquad G_6 = \tfrac{1}{7}x - u - \tfrac{12}{77}y + \tfrac{9}{77}z_4,\\ G_7 = w_0 - \tfrac{3}{2}a_0, \quad G_8 = w_5 - \tfrac{3}{2}a_5, \quad G_9 = w_7 - \tfrac{3}{2}a_7.\end{gathered}$$

The first four follow from $NHZ_P \geq 0$; the next two follow from (7.15) applied to (N, L) and (N, N), respectively. Use the order $G_D, G_A, G_B, G_T, G_5, \ldots, G_9$. Define α_j, μ_i by the following integer arrays, divided by 10^4:

$$10^4(\alpha_1, \ldots, \alpha_{25}) = \begin{array}{rrrrr} 125599 & -26317 & 838943 & -156924 & -2996 \\ -26435 & -17943 & -8274 & 4386 & 0 \\ -1024187 & -1649243 & 13946 & 14520 & 14040 \\ 17758 & 17386 & 31752 & 18397 & 26604 \\ -106826 & -49999 & 26631 & -16462 & -1519032 \end{array},$$

where the entries are read successively along the rows, and

$$10^4(\mu_1, \ldots, \mu_9) = \begin{array}{rrr} 73215 & 46413 & 8681 \\ 38908 & 6326668 & 1136405 \\ 10462 & 157397 & 79931 \end{array},$$

with the same row order. Put $Q = \sum_i \mu_i G_i - \sum_j \alpha_j E_j \geq 0$. Expansion gives constant term $-2003/2000$. Among its global variables, the positive coefficients of z_P, h_P, j_P have sum at most $29/30000$, and those of a_k, w_k sum to $437/320000$. The only other positive terms are $13x/14000 + 2y/9625$. Thus (7.16) bounds the total positive global contribution by

$$64\frac{29}{30000} + 160\frac{437}{320000} + \frac{1}{880} = \frac{5806}{20625} < \frac{3}{10}.$$

For a local tuple b, put $s = \sum b_i$, $v = \prod b_i$, and $\Lambda = \prod (b_i - 1)/v$ for an interior point, or $\Lambda = 1$ for a cusp. Its coefficient in Q is

$$\omega_\varepsilon(b, \gamma) = \alpha_{12}\Lambda + \sum_{k=0}^{7} c_k\big(s - \varepsilon(d_{K_k} + 1)\big)\gamma_k, \qquad c_k = \alpha_{13+k}.$$

The seven local equations give a common bound for this expression. Set

$$\begin{gathered} (r_0, r_1, r_2) = 10^{-4}(173706, -35568, 1857), \\ U = r_0 s - 191\varepsilon, \qquad V = r_1 s^2 + \tfrac{213}{2}\varepsilon s - 536\varepsilon^2, \qquad W = r_2 s^2 - 7\varepsilon s + 39\varepsilon^2, \\ \zeta_P = (r_1 \lambda_P + r_2 c_P)s + \varepsilon\eta_P, \qquad (\eta_D, \eta_A, \eta_B, \eta_T) = (19/2, 9/2, -19/5, 7/5). \end{gathered}$$

Direct substitution in the four surface rows and eight curve rows gives

$$\begin{gathered} U\lambda_{K_k} + \sum_P \zeta_P v_{P,K_k} \geq c_k(s - \varepsilon(d_{K_k} + 1)), \\ V\lambda_P + W c_P \geq \zeta_P(s - \varepsilon(d_P + 2)). \end{gathered}$$

For $\varepsilon = 1$, both differences are affine in s; the quadratic terms in the second cancel. It suffices to substitute $s = 8, 64$, where their respective minima are $687/5000$ and $39/125$. For $\varepsilon = 0$, the second difference is zero and the first is s times a positive coefficient, whose minimum over the eight rows is $1/80000$. Multiply the first inequalities by γ_k, apply the four incidence equations, multiply the second by β_P, and apply the remaining three equations. This proves

$$\omega_\varepsilon \leq \alpha_{12}\Lambda + \frac{Ue_3(b - \varepsilon) + Ve_2(b - \varepsilon) + 4W(s - 4\varepsilon)(s - 5\varepsilon)}{v}.$$

For the eighteen interior degree tuples other than F_4, the fraction divided by Λ is at most $2075924/13125 < -\alpha_{12}$. For the eight cusp tuples its maximum is $6597/40$, giving $\omega_0 \leq 7/10000$. At F_4, the support condition and the seven equations give uniquely

$$(\beta_D, \beta_A, \beta_B, \beta_T) = (1/4, 1/6, 1/8, 0), \qquad \gamma_1 = \gamma_4 = 1, \quad \gamma_k = 0 \ (k \neq 1, 4).$$

Substitution gives the common bound

$$\max \frac{\omega_\varepsilon(b,\gamma)}{\Lambda} = \frac{493}{154000} < \frac{1}{250}. \tag{7.17}$$

Thus the local comparison uses the same inequalities for every tuple, with one direct calculation at F_4. The displayed substitutions and the endpoint inequalities give the bound without enumerating vertices. Consequently the positive local contribution is at most $F_0/250 < 13/250$. We obtain

$$0 \le Q \le -1 + \frac{3}{10} + \frac{13}{250} = -\frac{81}{125} < 0,$$

a contradiction. □

Theorem 7.5. The hypothetical Gaussian arithmetic ball quotient with coarse compactification X_7 does not exist.

Proof. Propositions 6.2, 6.3, and 6.10 give all hypotheses of Lemma 7.4, in the same original group. The comparison uses neither integrality of L nor the exclusion of a root norm, and imposes no index or degree bound on an individual fixed component. □

A Local calculations for the Gaussian case

Lemma A.1. The two formulas and every local input in Lemma 6.5 use the full fixed normalizers. In its second and third rows, the effective surface lattice has exactly one cusp, with the holonomy and the positive elliptic line bundle stated there.

Proof. Substitute axes $(u{+}x, u{-}x, -u{+}y, -u{-}y)$, $(u{+}x, u{-}x, y)$ into the basic invariants, with frames (e_+, ue_-), $(u^2 e_{x^2}, e_{y^2})$ for A_3, T_3. In the order of the six groups in Lemma 6.5, determinant orders $(2,3,3,1,1,0)$, corrections $(-1/2,0,-1/2,0,-1/2,0)$, and inertia $(2,2,4,2,4,2)$ give both formulas. The full parabolic list [10, Tables 5, 6, 8 and Theorem 3.6] leaves finite types $T_2 \times A_1^2, T_3 \times A_1, T_2^2$, with normalizers $\mu_4 \times \mu_2, G(4,1,2), \mu_4^2$, inertia $2,4,4$, intersections $1,2,0$. At A_3, $(x, x, t{-}x, t{-}x)$, $t \in E[2]$, maps to a conic in $|4[0]|$, excluding a smooth surface germ; exchanges at B_3 give forbidden curves. For different coordinate labels, $(x_j^2) = (U, T, T{-}U, T{-}cU)$ gives full rings $\mathbb{C}[u,v]$ and

ambient degrees	D_P
1122	$(u^2-v^2)(cu^2-v^2)$
1222	$(v-u^2)(cv-u^2)$
1111	$(u-v)(cu-v)$

.

Only $2222/\mu_2$ fits row three: its two lines meet at one cusp, with ambient period, order 2, and bundle degree 2; none fits row two. The full quartic phases likewise leave different coordinate types; for $D_4 = q_{\mathrm{amb}}^2 x_2^4$, row one gives

parameter	C	D_Q	D_R
221	$iq_{\mathrm{amb}}y$	$D_4(D_4 - C^2)$	$\varnothing$
222	$-q_{\mathrm{amb}}^2 y^2$	$D_4(D_4 - C)$	$C.$

These are the required tacnode and node; row three forbids quartic supports, and row two excludes 212 by intersection $4 > 2$, 221 by contact 2. For 222, supports $(a_0, a_1, t), (a_0, r, t)$, $a_0 = (0,1,i), a_1 = (1,0,1), r = (1,1,0)$, give pairs $(D_4(D_4 - C), C), (D_4 - B^2, D_4)$, each exhausting the intersection and hence giving exactly one cusp. The restored phases give both scalars $(-1)^{b_1+a_2+u}$ for the first support, trivial scalar by $a_2 = b_2, b_1 + b_2 + u = 0 \bmod 2$ for the second. Thus $q_Z = q_{\mathrm{amb}}^2, q_{\mathrm{amb}}$, respectively; full holonomy μ_4 and actual degrees $(1,1),(1,2)$ give bundle degrees $4,2$, with no shorter period under free deck corrections. □

Lemma A.2. A Gaussian Picard lattice with orbifold Euler characteristic 3 lies in a maximal lattice of index 96. The maximal lattice has one cusp and horizontal order 4. The positive line bundle on its elliptic curve has degree 4 or 2.

Proof. Use the volume and normalizer formulas [26, Sections 3.3–4.1], [19, Section 2.3]:

$$3 \le n \le 3^{1+|I|}, \qquad \chi_{\mathrm{orb}}(\mathbb{B}^2/\Delta) = 3A/(32n), \qquad mA = 32n,$$

Here A is the local product, I the split Iwahori primes, n the full real index, $m = [\Delta : \Pi]$; center μ_3 and covolume $-L_{\mathbb{Q}(i)}(-2)/48 = 1/96$ give the displayed normalization. The nonminimal factors $(p^2+p+1)(p+1)$, $p \ge 5$, p^2-p+1, $p \ge 3$, force $I = \varnothing, A = 1, n = 3$. Class number 1 [19, Proposition 5.3] leaves the two ramified vertices, with Euler number $1/32$, index 96, one cusp [27, Theorem 4.8], and models

$$J = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad M_0 = \mathbb{Z}[i]^3,$$
$$M_s = \mathbb{Z}[i]e_1 + \mathbb{Z}[i]e_2 + \mathbb{Z}[i]\rho e_3, \qquad \rho = (-1+i)/2.$$

The vertex bases $((1-i)^{-1}e_1, e_2, \rho e_3), (e_1, e_2, \rho e_3)$ identify the standard group and its sister by Gram reduction modulo $1-i$ [29, Section 3.11]. For

$$U(z,t) = \begin{pmatrix} 1 & -\bar{z} & (-|z|^2+it)/2 \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix},$$

the full Heisenberg lattices are

$$N_0 = \{U(z,t) : z \in (1-i)\mathbb{Z}[i],\ t \in 2\mathbb{Z}\},$$
$$N_s = \{U(z,2a) : z \in (1-i)\mathbb{Z}[i],\ a \equiv |z|^2/2 \bmod 2\}.$$

The full groups are $N_0 \rtimes \mu_4, N_s \rtimes \mu_4$, with periods $2, 4$, area 2; commutator $U(0, 4\,\mathrm{Im}(z\bar{w}))$ gives degrees $4 \cdot 2/t_0 = 4, 2$. □

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